CHRISLAND UNIVERSITY, ABEOKUTA

COLLEGE OF NATURAL AND APPLIED SCIENCES

DEPARTMENT OF COMPUTER SCIENCES

STUDENT HANDBOOK

Programme: Software Engineering

DEPARTMENT OF COMPUTER SCIENCES
STUDENT HANDBOOK FOR SOFTWARE ENGINEERING

As a student in the Department of Computer Sciences, Chrisland University, Abeokuta, I understand that I must follow the procedures and policies that are included in this handbook as well as the policies of the University. I also undertake to study this handbook and seek clarifications on areas that I do not understand. It is my avowed commitment to be responsible for this booklet.

Student's Name: _____

Matriculation Number: _____

Signature: _____

Date: _____

This page is the property of Chrisland University and shall remain in the files of the Department of Computer Sciences.

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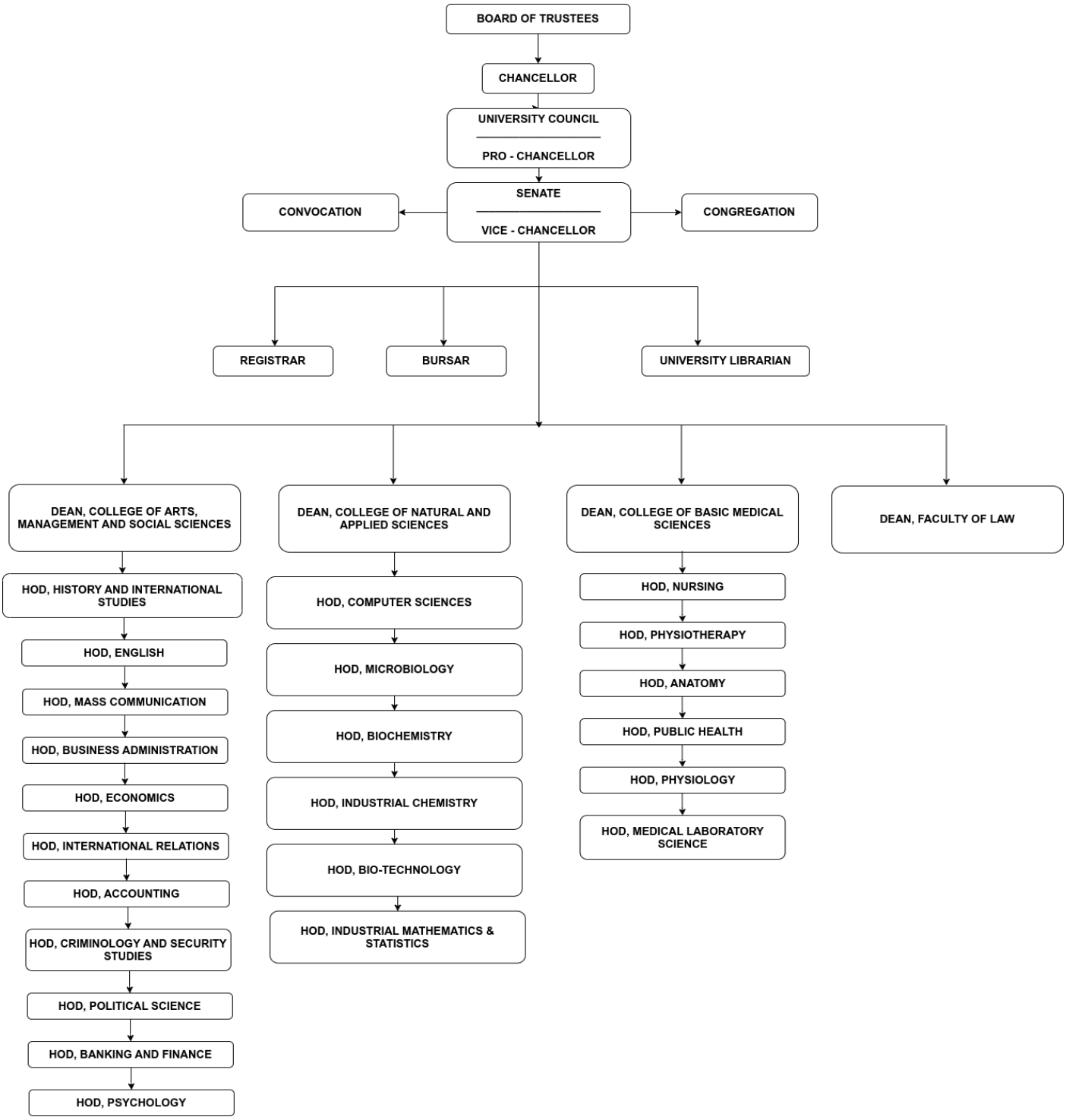
UNIVERSITY ORGANIZATIONAL STRUCTURE



UNIVERSITY MANAGEMENT STRUCTURE

Proprietor/Founder/Chairman Board of Trustees

High Chief Dr. Winifred Adefolahan Awosika (O.O.N.)

Chancellor

High Chief Dr. Winifred Adefolahan Awosika (O.O.N.)

Pro-Chancellor and Chairman of Governing Council

Distinguished Professor Ayodeji Oladimeji Olukoju FNAL

B.A., M.A., PhD

Vice Chancellor and Chairman of Senate

Professor Chinedum Peace Babalola

(Ph.D., B. Pharm, FAAS, FAS, FPSN, FPCPPharm., FNAPharm.)

Registrar

Mr. Samuel OMOTOSHO

University Librarian

Dr. Samuel Olukunle ADEYOYIN

Ag. Bursar

Mr. Moses ESAH

FOREWARD

Chrisland University is a vision-driven private university approved by the Federal Government. The take-off site is located in Abeokuta, Ogun State with the main campus presently under construction at Owode, Ogun State. The operating license from the Federal Government of Nigeria was obtained from the National University Commission in March 2015. The university's mission is to create a learning environment that nurtures a sound, moral and intellectual culture that attracts an excellent blend of faculty from across the world and deploys resources to educate the total person.

The university is created and operates with a purpose to assist in bridging the widening gap between potential student population and limited opening in the Nigerian tertiary educational system. Therefore the university hopes to attract talented youths, to teach and inspire them to advance the realm of knowledge by cultivating their faculty of academic enquiry, creative scholarships, research discovery and the application of the new knowledge to the transformation of the individual and the society.

In addition, the university has a vision to become a world class institution renowned for intellectual freedom, ethical standards, research, community service and the outstanding training of new generation of leaders. The curriculum supports an Entrepreneurial Skill Acquisition Scheme for job creation and self-employment at graduation.

Therefore, all qualified students are welcome to the university irrespective of colour, race, religion, tribe, or ethnicity. Welcome to Chrisland!!

PREFACE

This pamphlet contains information about the academic activities in the Department of Computer Sciences, Chrisland University, Abeokuta, Nigeria. The handbook is made available to both staff and undergraduate students in the Software Engineering Programme, as a general guide for both lectures and practical courses with a view to earning a Bachelor of Science (BSc.) Honours Degree in Software Engineering of Chrisland University.

It is our view that students will find the Departmental Handbook an indispensable and veritable tool for the fulfillment of their academic pursuit in the Department of Computer Sciences. The members of staff of the department welcome all students and wish them a very successful and productive stay in the Department and the in the University.

Professor Isaac A. Odun-Ayo
Head of Department
January, 2023

1.0 GENERAL INFORMATION

1.1 VISION

The vision of the department is;

- To be the leading Department in Research and Teaching towards the economic and developmental growth of the nation through ICT.

1.2 MISSION STATEMENT

To;

- Stimulate sustained interest and enthusiasm in Software Engineering;
- Train students the necessary computing knowledge and skills required to provide effective and timeless computer service;
- Train students with a solid background in Software Engineering and knowledge that are required for post graduate studies and research;
- Conduct research to advance the state of the art in Software Engineering and integrate its innovations into other disciplines.

1.3 PHILOSOPHY OF THE UNIVERSITY

The philosophy of Chrisland University is to provide a first class academic and research environment where the needs of the society and mankind in general are paramount to the achievements of the University.

Its philosophy is informed in the need to:

i. Create an atmosphere of academic freedom for deep exploration of mind and matter as well as the expansion of the frontiers of knowledge for proffering solutions to life's ultimate issues both for the individual and the society, and for advancing the cause of human happiness (emotional well-being), good health, environmental awareness, wealth creation and spiritual growth;

ii. Nurture a culture of research that defeats the age long practice and reliance on opinions, prejudice and anti-intellectual tendencies that dominate the public sphere in many developing societies, thereby dimming the light of development;

iii. Provide a liberal education that makes young people better prepared, spiritually strong, intellectually capable, and professionally competent to operate with distinction in most professional callings;

iv. Build up a new generation of knowledge and wealth creators so as to reverse the depth of poverty through the extension of opportunities, entrepreneurial skills, job creation, value re-orientation and the addition of value to all aspects of individual and corporate life; and

v. Foster a pragmatic link between the University and the world at large through citizenship training, community service and relentless advancement of human progress in all dimensions of life.

1.4 RESUMPTION DATES

The University, prior to the end of each semester/session or public/work free days will usually announce resumption dates as appropriate. Students are to comply with the schedule of resumption as may be published or announced by the University from time to time. Failure to resume on schedule will attract appropriate sanction.

1.5 ORIENTATION FOR FRESH STUDENTS

Fresh Students or *Freshers* must attend the compulsory Orientation Programme.

i. The Orientation Programme which holds at the beginning of a new session is intended:
   a. to enable new students to settle down and adjust to the University environment, and
   b. To afford students the opportunity to familiarize themselves with the existing rules, regulations and procedures of the University.

ii. The Students Affairs Unit in collaboration with other departments organizes the Orientation Programme.

iii. It is mandatory that all freshmen/women attend all the orientation activities and sign the attendance register.

iv. Any student who has received a letter of admission before the orientation week, but fails to attend any of the orientation week activities may have his/her admission withdrawn.

2.0 UNIVERSITY FACILITIES AND STUDENTSHIP

2.1 UNIVERSITY HEALTH SERVICES

i. Every student must register with the University Health Centre as part of the registration exercise.
ii. For any form of ailment, students should report at the University Health Centre for treatment or advice.
iii. For official purposes only the report of the Director of the University Health Centre is acceptable to the University.
iv. Even for treatment received outside the University, the medical report, if required for official purposes, must be endorsed by the Director of the University Health Centre.
v. The University must be informed of any sickness of students, outside the University during the session.

2.2 THE UNIVERSITY LIBRARY

As part of the registration exercise, every student must register at the University Library and must abide by the following rules:

i. Library users are expected to show respect for the rights of others by keeping quiet in the library and provide an enabling environment for study and research. Therefore, all handsets are required to be put off or kept on silent mode while in the library. No call should be received in the library.
ii. No library material should be taken out of the building without signing for it at the circulation desk.
iii. All library materials borrowed (issued) are to be returned promptly.
iv. Library materials issued are to be treated with proper care by whom used in the library.
v. No food or drink is allowed in the library.
vi. All readers are to surrender their bags and books for inspection at the security post when leaving the library.
vii. All library users must show library identity cards before entry is granted into the library.
viii. While entering the library, all personal belonging except purses, note books, private books and books meant to be returned to the library, shall be deposited by each reader at the property counter.

In addition, students should note that any act of misconduct in the library, will attract appropriate sanctions.

2.2.1 THE COLLEGE LIBRARY

All of the rules and regulations stated above for the University Library holds for the departmental library. Please take note.

2.3 REGISTRATION OF STUDENTS AND PROCEDURE FOR REGISTRATION

i. All freshmen students must be cleared in sequence by the Admission Office, Bursary and Student Affairs Unit on first arrival.
ii. All Students must register at the beginning of each Semester for the course prescribed by their Colleges.
iii. Registration must be completed within the prescribed period. Beyond that period, late registration may be permitted up to a limit and on the payment of a later registration fee determined by the Senate.
iv. All registration (whether late or completed at regulation time) will stop on a specific date, prior to matriculation.
v. On no account will registration be permitted after matriculation.
vi. The step-by-step registration process is as follows:
    a. Clearance by the Admissions Officer and collection of registration forms from the Academic Office.
    b. Payment of prescribed fees and clearance from the Bursary.
    c. Allocation of students into Halls of residence for only students who have paid their fees.
    d. Clearance by the Faculty Office.
    e. Completion of registration forms in consultation with Heads of Departments concerned
    f. Submission of completed registration forms duly signed and forwarded to the Academic Office through the College Officer.
    g. During the period of registration, all fresh students must register at the University Health Centre.
    h. A copy of the final completed registration form must be collected from the College Officer to be kept by the student.

2.4 MATRICULATION

i. All freshman students, properly registered as indicated in section **2.3**, shall be formally admitted into the University at a Matriculation Ceremony to be conducted on a date specified by the Senate.

ii. It is compulsory for all freshman students to take the Matriculation Oath to be administered on the Matriculation Day.

iii. A student becomes a bonafide student of the University only after taking the Matriculation Oath, signing the Matriculation Register and being allotted Matriculation number.

iv. For emphasis, no registration is allowed for any student after the Matriculation day.

2.5 GUIDELINES FOR BEHAVIOUR ON CAMPUS

i. Any student who display any form of disrespect bordering on contempt, to any official of is performing his/her official assignment will be subject to disciplinary action.

Any student who writes an insulting letter to an official of the University will be subject to disciplinary action.

2.6 DRESS CODE

i. All students must dress corporate except when they are in the Halls of residence or actively participating in sporting activities or field practice.

ii. While maintaining good taste with high moral and modesty, the appearance of every student should be characterized by neatness, purity and simplicity, which must conform to the unique standards and virtues of Chrisland University.

2.6.1 CAMPUS DRESS CODE FOR FEMALE STUDENTS

The following regulations apply at all times to all female dressing on campus.

i. Dressing which is sexually provocative or expose vital parts of the body that are supposed to be concealed (e.g. chest, navel and thigh) is prohibited.

ii. The identity of all students must be visible at all times (i.e. faces must be fully visible); the forms of dressing that obscures easy identification are not allowed.

iii. Skirt must be long enough to cover the knees and must not be slit above the knees.

iv. Transparent, sheer, or tight fitting blouses, sweaters or skirts are not permitted.

v. Dresses with low necklines or dresses that do not fully cover the shoulders and armpits are not permitted.
vi. Shorts, slacks and similar attire should not be worn outside the residence Halls on campus, except while on field trips.
vii. Trouser suits may be worn but with camisole that are not with low necklines.
viii. Excessive facial make-up, multi coloured nail-polish, artificial nails and bleaching are not allowed.
ix. Students are allowed moderate use of unnatural braided hair or attachment, which in any case should not exceed neck-length.
x. Unnatural braided hair or attachment should match with the natural hair colour.
xi. The hair should be well groomed (neat and tidy) all the time.
xii. Use of jewelry should be moderate.
xiii. Noisy and bogus shoes should be avoided.

2.6.2 CAMPUS DRESS CODE FOR MALE STUDENTS

i. Men's attire for lectures/ and University events should consist of trousers (not jeans), shirt and tie to fit.
ii. Men should not go bare-footed or without shirts anywhere on campus and must wear proper sports uniforms when participating in sports activities or outdoor game.
iii. Men should not wear muscle tops, earrings or necklaces.

3.0 BRIEF HISTORY OF THE DEPARTMENT

The University at inception in 2015 created two colleges which are; College of Management and Social Sciences (COMASS) and College of Natural and Applied Sciences (CNAS). CNAS started with four departments. Software Engineering programme was an arm of the Department of Physical Sciences established in 2015. During these years, Software Engineering programme under the Department of Computer Science has evolved to become a unique academic division in the production of highly trained students in the field of Computer Science. In addition, the Department offers the programme leading to the award of Bachelors of Software Engineering degree in Computer Science.

The department has enjoyed some increment in student population though this is a major goal at this point. Lectures are delivered to the students in the assigned lecture halls on campus. The structure of the program meets the National Universities Commission (NUC)'s Benchmark for Minimum Academic Standards. It also complies with global requirements for a degree in Software Engineering.

Staff members of the Department are in the forefront in the recent areas of computer research work. This has enabled her to continue training quality students who will be able to occupy various positions in banks, industries, academia and private companies.

3.1 PHILOSOPHY AND OBJECTIVES OF THE DEPARTMENT

The Philosophy and Mission Statement underlying the programmes in Software Engineering is aimed at achieving the goals and objectives of the National Policy on Industrialisation and Self-Reliance. This is to be achieved through:

i. Broad-based foundation in Computing, Science and Engineering and Technology as well as specialized knowledge and practice in Software Engineering.
ii. Practical exposure to application of Software Engineering to problem solving.
iii. Adequate training in human and organizational behavior and management in the software development life-cycle.
iv. Developing in the products entrepreneurial knowledge, a sense of public responsibility and a spirit of self-reliance.

v. Nurturing of partnership between the institution and the software industry for effective programme delivery.
vi. Creating an awareness and understanding of the moral, ethical, legal and professional obligations needed to function as part of the computing ecosystem while protecting human and welfare and the environment in a global society.
vii. Creating an awareness and understanding of the need to develop leadership and team building skills to maximize the benefits of Software Engineering education and its application to solving problems.

The general philosophy therefore is to produce graduates with high academic and ethical standards and adequate practical exposure for self-employment as well as being of immediate value to the software industry and the community in general.

3.1.1 THE OBJECTIVES OF THE B.SC. DEGREE IN SOFTWARE ENGINEERING ARE TO:

The general goal and objectives of Software Engineering education and training should be in consonance with the realisation of national needs and aspirations vis-à-vis industrial development and technological emancipation. The graduates must therefore be resourceful, creative, knowledgeable and to perform the following functions.

(a) To equip students with deep understanding and wide-ranging knowledge of technical and developmental issues associated with software engineering practice;
(b) To develop skills required to use a range of software engineering tools and techniques in the development of robust and effective software systems across a range of domains.
(c) To provide and promote sound practicals and theoretical training in software engineering that will make the students useful in both private and public sectors of the economy;
(d) To equip the students with the capacity for professional practice in computing and communications;
(e) To develop in students through an education in software engineering a range of transferrable applicable skills to all aspects of human endeavors;
(f) To develop the students for the purpose of self-employment and job placements in the government and industry.
(g) To generate in students the ability to appreciate the importance of software engineering in an industrial, economic technological and social context; and
(h) To develop the students through sound academic and professional programmes as a basis for further studies in software engineering or multi-disciplinary studies.

3.2 VISION AND MISSION STATEMENT OF THE DEPARTMENT

The vision of the Department is:

- To be the leading Department in Research and Teaching towards the Economic Growth of the Nation through ICT.

The mission of the Department is to:

- Stimulate sustained interest and enthusiasm in Software Engineering.
- Train students the necessary computing knowledge and skills required to provide effective and timeless computing service.
- Train students with a solid foundation in Software Engineering and knowledge that are required for post graduate studies and research.
- Conduct Research to advance the state of the art in Software Engineering and integrate its innovations into other disciplines.

4.0 ADMISSION REQUIREMENTS

4.1 ACADEMIC REGULATIONS GOVERNING THE AWARD OF BACHELOR'S DEGREE

4.2 THE COURSE SYSTEM

Chrisland University, Abeokuta operates a Course system. This is a system in which subjects are broken down into one or more convenient sections called Course, which are taught and examined within a semester. A Course is assigned a specific number of lectures and practical/tutorial hours and the total number assigned to it will determine its unit value. A Course unit is defined as fifteen hours of lecture (one lecture per week) or forty-five hours of practical work (3 hours of practical work per week) for 15 weeks. Regulations guiding the operation of the Course system in the College of Natural and Applied Sciences are specified in the prospectus.

The University also operates the Semester system under which full academic session is divided into two semesters. A full semester consists of 15 weeks of lectures plus practical, after which examinations are taken.

4.3 COURSE CREDIT SYSTEM

This should be understood to mean a 'quantitative system of organization of the curriculum in which subject areas are broken down into unit courses which are examinable and for which students earn credit(s) if passed'. The courses are arranged in progressive order of difficulty or in levels of academic progress, e.g. Level or Year I courses are 100,101 etc., and Level II or Year II courses are 200, 202 etc.

The second aspect of the system is that courses are assigned weights called Credit Units.

4.3.1 Grade Point, Grade Point Average and Cumulative Grade Point Average

As there is considerable variation in the modes of application and interpretation of Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA) as means of computing mean score of a student, there is, therefore, the need to define and streamline their use as follows:

(i) Grade Point (GP):

The Grade Point derives from the actual percentages, raw score for a given course; the raw score is converted into a letter grade and a grade point.

(ii) Grade Point Average (GPA):

Performance in any semester is reported in Grade Point Average. This is the average of weighted grade points earned in the course taken during the semester. The Grade Point Average is obtained by multiplying the Grade Point Average in each course by the number of Credit Units assigned to the course, and then summing these up and dividing by the total number of Credit Units taken for the semester.

(iii) Cumulative Grade Point Average (CGPA)

This is the up-to-date mean of the Grade Points earned by the student in a programme of study. It is an indication of the student's overall performance at any point in the training programme. To compute the Cumulative Grade Point Average, the total of Grade Points multiplied by the respective Credit Units for all the semester are added and the divided by the total number of Credits Units for all courses registered by the student.

4.4 REGULATIONS GOVERNING THE OPERATIONS OF THE COURSE SYSTEM

i. Chrisland University, Abeokuta shall adopt the Course System in conducting all its academic programmes.
ii. A course is any part or whole of a subject which can be completed through lectures, tutorials and/or practical with a specified number of hours within a semester.
iii. Courses are normally weighed in terms of Course Units ranging from one to six.
iv. A Course is defined as subject matter completed in fifteen lecture hours or forty-five practical hours. Whenever tutorials are included in the instruction, they are rated as lecture.

4.5. Undergraduate Courses shall be allotted the following level code:

101 – 199 for 1st year students

201 – 299 for 2nd year students

301 – 399 for 3rd year students

401 – 499 for 4th year students

For entry into Bachelor of Science degree programme in Software Engineering candidate must satisfy the University's minimum admission requirements spelt out in this Handbook. The specific requirements include:

TABLE FOR REQUIREMENT ENTRY INTO SOFTWARE ENGINEERING

Course	**Direct Entry Requirements**	**O'Level Requirements**	**UTME Subjects**	**Waivers and Special Consideration/Other Qualifications**
Software Engineering	NCE in Software Engineering with any of Mathematics, Chemistry, Physics, Integrated Science, Economics or Two 'A' Level Passes in Physics and Mathematics or Chemistry.	Five 'O' Level SSCE/WASC Credit Passes in English Language, Mathematics, Physics, Chemistry plus any other science subjects.	Mathematics, Physics and Chemistry	Holders of OND/NCE (minimum upper credit) in relevant subjects may be considered for Direct Entry to 200 Level provided they satisfy the UTME requirements.

4.6 COURSE CATEGORIES AND CREDIT SYSTEM

Chrisland University operates a course system. This is a system in which subjects are broken down into one or more convenient sections called Courses, which are taught and examined within a semester. A course is assigned a specific number of lecture and/or practical hours and total number of hours assigned to it will determine its unit value. The University also operates the semester system under which a full academic session is divided into two semesters. A full semester consists of 15 weeks of lectures plus practical, after which examination are taken.

a. A course may be evaluated in terms of lecture hours alone without tutorial or practical hours, but may also be a combination of lecture, tutorial and practical hours.

b. For all undergraduate programmes, the following are the general categories of courses:

   (i) **Core/Compulsory**: Courses so categorized must be taken and passed before any student can graduate in any degree programme. For the entire University, they are termed core courses. They are designated for each degree programme and not subject to any choice by students.

   (ii) **Elective**: Course designated as elective are left for students to choose to make up their work load or degree programmes. They may be from their field of study of outside them, but should be relevant to their programmes prescribed minimum number of units from such courses must also be passed before the student can graduate.

c. A course must be completed and examined within a semester and students shall be credited with the appropriate number of course units assigned to each course passed.

d. The pass mark for all courses examined shall be 40 percent (40%).

e. **Workload**: The minimum number of course units for which a student can register in any semester shall be 16 units while the maximum shall be 24. Any student wishing to register for less than the minimum or more than the maximum shall seek permission of Senate through his/her College Board.

f. **Probation and Withdrawals**: Any students who scores a Grade Point Average (GPA) of less than 1.50, at the end of one semester shall be places on probation during the next semester.

g. Any student who has a CGPA of less than 1.50, at the end of an academic session shall be on probation

h. Any student that scores a CGPA of less than 1.50, in two consecutive semesters within an academic session shall be required to withdraw from the programme.

i. A student who has a GPA of less than 1.50, in two consecutive semesters within a session and a CGPA of less than 1.00 (one), at the end of the session shall be required to withdraw from the University.

4.7 CLASSIFICATION OF DEGREES AND COURSE GRADES

All letter grades are assigned specific grade points ranging from 5 (for A) to 0 (for F), and these grade points are used in the computation of GPA and CGPA which form the basis for degree classification at the end of each academic programme as shown in the table below.

All prescribed courses are considered in the computation of the final CGPA for the degree classification at the Chrisland University.

4.8 WORKLOAD

1. The minimum course load for which a student can register in any semester shall be 15 units while the maximum shall be 24 units. Any student wishing to register for less than the minimum or more than the maximum shall seek permission of Senate through his/her College Board.
2. Exception: Students in the 200-400 levels of their Programme may be allowed to register for a maximum of six additional units per session (three units per semester). Approval shall be given by the College Board which shall report to the Senate for ratification.

4.9 PROBATION AND WITHDRAWALS

Any student who scores a grade point average (GPA) of less than 1.50 at the end of one semester shall be placed on probation during the next semester. Any student that scores a GPA of less than 1.50 in two consecutive semesters within an academic session shall be required to withdraw from the University.

Any student asked to withdraw from a Programme shall be allowed to transfer to another Programme provided he/she has the minimum requirement to be admitted into the new Programme.

4.10 TRANSFERS TO A NEW PROGRAMME

The highest level for transfer to a new Programme is at 200 level depending on the similarities in the course combination for the degree Programme. Whichever level he/she transfers into, it shall be assumed that he/she is starting a new degree Programme afresh and all his/her past records (except those of 100 level which are common to all degree Programme) will not count in the computation of his/her degree classification, though, they will appear in the academic transcripts for complete work.

NOTE: The maximum number of times a student can be allowed to transfer to a new Programme is once.

4.11 TRANSFER OF STUDENT FROM OTHER UNIVERSITIES

Students from other universities who wish to transfer to Chrisland University, Abeokuta:

i. Must apply for the appropriate transfer form and pay the prescribed fee.
ii. Must complete the forms and send them along with his/her academic transcripts. A confidential report from the Registrar of his/her University must be sent to the Registrar of Chrisland University within the stipulated time allowed before admission exercises begin.
iii. May be conditionally admitted into any of the levels but in any case not higher than 200 levels in the degree Programme. Such may in addition have some lower level course, prescribed as remedial course which must be passed before graduation: and
iv. Shall have all academic records brought from their University excluded from the computation of the results to be used in the award of degree.

4.12 ADDITION AND DELETION OF COURSE

i. There shall be no re-sit examination in any course.
ii. When a student scores below the pass mark 40 percent in any course, he/she shall be required to carry the course over into the next available semester when the course is offered.
iii. All carry over courses must first be registered for before any additional course are registered for.

iv. Whenever the total number of carry over courses exceed the minimum work load of 16 for that semester, such a student will not be allowed to proceed to higher course until he/she passes those courses even if the CGPA exceeds 1.5.

There is no limit to the number of times a student can retake a course examination being carried over but all marks scored in all examination attempts shall appear in his/her examination record until he/she passes the examination and they shall all count in the final determination of his CGPA and therefore his/her class of degree.

4.13 REGISTRATION OF COURSES

1. The prescribed set of course for each year of every degree programme as approved by the Senate, shall be registered for at the beginning of each academic year on the approved course registration form under the guidance of the Head of Departments or designated staff.
2. No students shall register for more course units than maximum and less than the minimum course units prescribed by Senate without its permission.
3. Courses are considered registered for when the Heads of Department and the course lecturers or their representatives have endorsed each course on the registration form.
4. Courses not endorsed by the Head of Department, their representatives and the course lecturers or which do not appear at all on the registration forms shall be regarded as invalid, null and void, even if any student attends lecturers in those courses and pass the set examinations.
5. A student must attend at least 80% of all lectures and practical for any course registered for before he/she can be allowed to sit for the examination for such courses.
6. Any alteration on the course registration forms not endorsed and dated by the designated staff shall be rendered null and void and the mark scored invalid.
7. Any course or unit registered above the maximum unit allowed (24 units) without approval/permission in writing and attached to the course registration for the appropriate semester shall be rendered null and void and the mark due for the equal amount of unit above the limit shall be deducted from the number of course already passed.

4.14 LATE REGISTRATION OF COURSE

1. The normal period within which to complete all registration formalities shall be 2 weeks (or any duration as approval by Senate) from the date of commencement of the registration exercise.
2. Any registration formalities not completed within the first 2 weeks (or any duration approval by Senate) SHALL normally be registered as late and shall attract a penalty fee of N2,500 per week (or as may be determined from time to time) unless acceptable reasons are given for the lateness.
3. Registration of courses, however, shall not be allowed after the fourth week of commencement of lectures for those courses.
4. A student shall normally be allowed to add or delete courses from the list of courses already registered for, providing he/she is not violating the minimum and maximum workload registration.
5. Such request to add or delete courses shall normally be made on the prescribed form obtainable from the academic office.
6. Any student intending to delete or add course shall normally be expected to seek the advice of his/her Head of Department or Academic adviser before proceeding to complete the form.
7. All requests to add course must be made within the first four weeks of each semester in which the course are being offered.
8. Request to delete course shall normally not be permitted beyond the end of the 6th week of the commencement of lectures in the course.

4.15 DURATION OF DEGREE COURSE

1. To qualify for an honors degree, a student shall complete his/her degree requirement during the minimum period prescribed or a period not exceeding three additional years beyond the prescribed minimum duration.
2. Any student who spends more than three extra years over and above the prescribed minimum shall only be awarded a pass degree.
3. Students who cannot meet all their degree requirements within three years in excess of the minimum duration shall be required to withdraw from the University without the award of a degree.

5.0 QUALIFYING FOR THE AWARD OF A DEGREE

To qualify for the award of a degree, a student must meet the following requirements:

i. Must have spent not less than the minimum period and not exceed the maximum duration prescribed for the programme (3 years over and above the minimum).

ii. Must have registered for and passed all CORE/COMPULSORY, REQUIRED and ELECTIVE courses prescribed for the degree Programme.

iii. Must have passed not less than minimum course unit specified for the degree.

iv. Must have obtained not less than a CGPA of 1.50 at the end of the programme

v. Must have satisfied all other University requirement as may be prescribed from time to time by Senate.

vi. Must also have been found worthy in character throughout the duration of his/her studies

5.1 COURSE EVALUATION

Course shall be evaluated on the basis of continuous assessment and examination conducted at the end of each course.

Courses which are largely lecture or tutorial based shall be evaluated on the basis of continuous assessment score of 30 percent (short tests, take-home assignments, term papers etc.) and course examinations of 70 percent conducted at the end of the semester.

Courses which are largely practical based shall be evaluated by direct assessment basis (up to 80 – 90 percent) and a short exam not exceeding 10 – 20 percent of scores.

Courses based on Industrial Training shall be evaluated by direct assessment of student to performance in the industry, as well as student's report and seminar presentation or oral interview.

TABLE SHOWING SCORING AND GRADING SYSTEMS

Exam Score Percent	Letter Grade	Grade Point	CGPA	Class of Degree
70 – 100	A	5	4.50 – 5.00	1st Class
60 – 69	B	4	3.50 – 4.49	2nd Class (Upper)
50 – 59	C	3	2.40 – 3.49	2nd Class (Lower)
45 – 49	D	2	1.50 – 2.39	3rd Class
40 – 44	E	1	0.00 – 1.49	Fail
00 – 39	F	0	-	-

5.2 PURPOSE, AIMS AND OBJECTIVES OF THE DEGREE PROGRAMME

The purpose, aims and objectives of Bachelors honors degree programme in Software Engineering should include:

• To create in students the awareness of and enthusiasm for Software Engineering and its capabilities.

• To involve the students in an intellectually stimulating and satisfying experience of learning and studying.

• To provide a broad and balanced foundation in Software Engineering knowledge and practical skills.

• To develop in students through education in Software Engineering a range of applicable transferable skills of information technology to all aspects of human endeavors.

• To generate in students an appreciation of the importance of computer in an industrial, economic, technological and social context.

• To provide students with the knowledge and skills base for further studies in Software Engineering or multi-disciplinary studies involving Software Engineering.

5.3 ADMISSION AND GRADUATION REQUIREMENT

The entry requirements shall be at least credit level passes in five subjects including English Language, Mathematics, and Physics to form the core subjects with credit in any other two relevant science subjects, at the Senior Secondary School Certificate or its equivalent. In addition, an acceptable pass in the University. Candidates with two A-level passes (graded A-E) at the Advanced Level in one or more relevant subjects (Chemistry, Mathematics, and Physics) may undertake the three – year degree programme into 200-level.

Any student having A CGPA of less than 1.5 in two consecutive sessions shall be asked to **withdraw from the programme**.

Any student having a CGPA of less than 1.0 in two consecutive sessions shall be asked to **withdraw from the university**.

Graduation Requirement

1. All courses registered for must be passed.
2. All courses designated as compulsory must be taken and passed
3. All stipulated GST courses must be taken and passed.
4. For any student to graduate, he/she must have taken 144 units for UTME candidates and 108 units for direct entry candidates

5.4 LEARNING OUTCOMES

a) Regime of Subject Knowledge

Each institution providing degree programmes in Software Engineering is free, within the context of university autonomy and academic freedom to decide on the content, nature, and

organization of its courses and modules. However, it is expected that all programmes will ensure that students are conversant with the core areas of Software Engineering:

b) Competencies and Skills

Students are expected to develop a wide range of different abilities, dynamism, and skills.

These may be divided into three categories, viz.

Cognitive Abilities and Skills Practical Skills General Skills

c) Behavioral Attitudes

General skills relating to non-subject specific competencies, communication, interpersonal, organization skills.

d) Attainment Levels:

Graduates of Software Engineering are expected to have the ability to apply knowledge and skills to solving theoretical and practical problems in Software Engineering, development of relevant ICT for national development and societal needs.

BMAS

6.0 B. SC. SOFTWARE ENGINEERING COURSES

100 LEVEL FIRST SEMESTER

Course Code	Status	Course Title	Contact Hours			Unit(s)
			L	T	P	
CSC 101	C	Introduction to Computing and Applications	2	0	3	3
CHM 101	C	General Chemistry I	2	1	0	3
CHM 103	C	General Chemistry Laboratory I	0	0	3	1
GST 101	R	Communication in English I	1	1	0	2
GST 107	R	Use of Library, Study Skills and ICT	1	1	0	2
GST 111	C	Introduction to Music	1	0	0	0
MTH 101	C	Elementary Mathematics I	2	1	0	3
PHY 101	C	General Physics I	2	1	0	3
PHY 107	C	General Physics Laboratory I	0	0	0	1
		TOTAL				**18**

100 LEVEL SECOND SEMESTER

Course Code	Status	Course Title	Contact Hours			Unit(s)
			L	T	P	
SEG 102	C	Principles of Programming I	2	0	3	3
SEG 104	C	Introduction to Web Technologies	1	0	3	2
CHM102	C	General Chemistry II	2	1	0	3
CHM104	C	General Chemistry Laboratory II	0	0	3	1
GST 102	C	Communication in English II	2	0	0	2
GST 106	C	Logic, Philosophy and Human Existence	2	0	0	2
GST 110	C	Contemporary Health Issues	2	0	0	2
MTH 102	C	Elementary Mathematics II	2	1	0	3
PHY 102	C	General Physics II	2	1	0	3
PHY 108	C	General Physics Laboratory II	0	0	3	1
		TOTAL				**22**

200 LEVEL FIRST SEMESTER

Course Code	Status	Course Title	Contact Hours			Unit(s)
			L	T	P	
SEG 201	C	Introduction to Software Engineering	2	0	0	3
SEG 203	C	Web Applications Development	2	0	3	3
SEG 205	C	Software Requirements and Design	2	0	0	2
SEG 207	C	Principles of Operating Systems	2	0	3	2
MAT 205	C	Linear Algebra I	1	1	0	2
GST 205	C	Entrepreneurship Studies I	1	1	0	2
GST 207	C	Nigeria's People, Culture, Government and Economy	2	0	0	2
GST 209	C	Morality, Religion, Ethics and Etiquette	1	0	0	1
GST 211	C	Instrumentation in Music	1	0	0	0
		TOTAL				**17**

Electives: Elective courses may be selected from the following: CSC 203 Foundations of Sequential Program (3 units), CSC 201 Computer Programming I (3 units), MTH 211 Statistics for Physical and Chemical Sciences (3 units), MAT 201 Mathematical Methods I (3 units)

200 LEVEL SECOND SEMESTER

Course Code	Status	Course Title	Contact Hours			Unit(s)
			L	T	P	
CSC 204	C	Fundamentals of Data Structures	2	0	3	3
CSC 208	C	Discrete Structure	2	1	0	3
SEG 204	C	Principles of Programming II	2	0	3	3
SEG 206	C	Logic and Its Application in Computer Science	2	1	0	3
SEG 208	C	Software Engineering Process	2	0	0	2
SEG 202	C	Software Construction	1	0	3	2
GST 206	C	Entrepreneurship Studies II	2	0	0	2
GST 204	E	Peace and Conflict Resolution	1	0	0	1

GST 208	C	Communication in French I	1	0	0	1
		TOTAL				**20**

300 LEVEL FIRST SEMESTER

Course Code	**Status**	**Course Title**	**Contact Hours**			**Unit(s)**
			L	**T**	**P**	
CSC 301	C	Structured Programming	2	0	0	3
SEG 305	C	Software Testing and Quality Assurance	1	0	3	2
SEG 309	C	Computer Architecture and Organisation	2	1	0	3
SEG 311	C	Design and Analysis of Computer Algorithms	2	1	0	2
SEG 313	C	Data Structures and Algorithms	2	0	2	3
SEG 315	C	Object-Oriented Analysis and Design	2	0	3	3
MAT 315	C	Operation Research I	2	1	0	3
ENT 301	C	Entrepreneurship Studies III	2	0	1	2
		TOTAL				**21**

Electives: Students may wish to register for any from the under listed courses: SEG 301 Formal Methods in Software Development (2 units), SEG 303 Model-Based Software Engineering (2 units), SEG 307 Software Engineering II (3 units), CSC 309 Systems Analysis and Design (3 units), CSC 313 Object-Oriented Programming (3 units)

300 LEVEL SECOND SEMESTER

Course Code	**Course Title**	**Units**	**Status**
SEG 399	Student Industrial Work Experience Scheme (SIWES)	16	C
TOTAL		**16 Units**	

NB: Any student having a cumulative of 12 units outstanding for second semester courses is not eligible to go for SIWES.

400 LEVEL FIRST SEMESTER

Course Code	Status	Course Title	Contact Hours			Unit(s)
			L	T	P	
SEG 403	C	Software Engineering Professional	2	0	0	2
SEG 409	C	Human Computer Interaction	2	0	0	3
CSC 411	C	Computer Networks and Communications	2	0	3	2
SEG 401	C	Software Configuration Management and Maintenance	1	0	3	2
SEG 405	C	Concepts of Programming Languages	2	0	0	3
SEG 407	C	Research Methodology	1	0	0	1
SEG 413	C	Database Systems	2	0	0	3
LDS 401	C	Leadership Studies	1	0	0	1
		TOTAL				**17**

Electives: Students may wish to register the under listed course as elective: CSC 403 Net-Centric Computing (3 units)

400 LEVEL SECOND SEMESTER

Course Code	Status	Course Title	Contact Hours			Unit(s)
			L	T	P	
SEG 416	C	Software Engineering Security	1	1	0	2
SEG 414	C	Distributed, Parallel, and Cloud Computing Systems	2	0	3	2
SEG 404	C	Software Engineering Project Management	1	1	3	2
SEG 406	C	Software Engineering Economics	1	0	0	2
SEG 412	C	Open Source Development and Applications	1	0	3	2
SEG 418	C	Software Architecture and Design	2	1	0	2
SEG 499	C	Final Year Project	0	0	18	6
ENT 302	C	Entrepreneurship Skills	2	1	0	2
GST 308	C	Communication in French II	1	0	0	1
		TOTAL				**21**

ELECTIVE COURSES

Course Code	Status	Course Title	Contact Hours			Unit(s)
			L	T	P	
SEG 402	E	Engineering Mobile Applications	1	0	3	2
SEG 408	E	Software Engineering Models and Methods	2	0	0	2
SEG 420	E	Artificial Intelligence and Expert Systems	2	0	3	3
SEG 422	E	Special Topics in Software Engineering	2	0	0	2
SEG 424	E	Fault Tolerant Computing	1	0	3	2
SEG 426	E	Embedded Systems	2	0	0	2
SEG 428	E	Game Design and Development	2	0	0	2
SEG 430	E	Modelling and Computer Simulation	1	0	3	2

COURSE DESCRIPTION

CSC 101: INTRODUCTION TO COMPUTING AND APPLICATIONS (3 Units)
Introduce the Computer as a tool for content creation, storage, and processing. Some of the broad fields of Computer Sciences from Programming Languages to Artificial Intelligence. Computing methods to problem solving and programming. Emphasis is placed on gaining literacy and some practice with computers, applications and problem solving, and the broad areas of computing.

CHM 101: GENERAL CHEMISTRY 1 (3 Units)
Atoms, molecules, and chemical reactions. Modern electronic theory of atoms. Electronic configuration, periodicity and building up of the periodic table. Hybridization and shapes of simple molecules. Valence forces; structure of solids. Chemical equations and stoichiometry; chemical bonding and intermolecular forces, kinetic theory of matter, elementary thermochemistry; rates of reaction, equilibrium and thermodynamics. Acids, bases and salts. Properties of gases. Redox reactions and introduction to electrochemistry. Radioactivity

CHM 107: GENERAL PRACTICAL CHEMISTRY 1 (1 Unit)
Laboratory experiments designed to reflect the topics taught in CHM 102 and CHM 102 such as qualitative chemical analysis, acid-base titrations. Gravimetric analysis. Calculation, data analysis and presentation. Functional group analysis.

GST 101: COMMUNICATION IN ENGLISH 1 (2 Units)
Effective communication and writing in English Language skills, essay writing skills (organization and logical presentation of ideas, grammar and styles), Comprehension, Sentence construction, Outlines and paragraphs.

GST 107: USE OF LIBRARY, STUDY SKILLS AND ICT (2 Units)

Brief history of libraries; library and education; university libraries and other types of libraries; study skills (reference services);Types of library materials, using library resources including e-learning, e-materials, etc. ; understanding library catalogue (card, OPAC, etc.) and classification; copyright and its implications; database resources; bibliographic citations and referencing. Development of modern ICT; Hardware technology; software technology; input devices; storage devices; output devices; communication and internet services; word processing skills (typing, etc.)

MTH 101: ELEMENTARY MATHEMATICS I (Algebra and Trigonometry) (3 Units)

Elementary set theory, subsets, union, intersection, complements, and Venn diagrams. Real numbers; integers, rational and irrational numbers. mathematical induction, real sequences and series, theory of quadratic equations, binomial theorem, Complex numbers; algebra of complex numbers; the Argand diagram. De- Moires theorem, nth roots of unity, Circular measure, trigonometric functions of angles of any magnitude, functions of angles of any magnitude, addition and factor formulae.

MTH 102: ELEMENTARY MATHEMATICS II (3 Units) (Calculus)

Functions of a real variable, graphs, limits and idea of continuity. The derivative, as limit of rate of change. Techniques of differentiation, maxima and minima. Extreme curve sketching, integration, definite integrals, reduction formulae, application to areas, volumes (including approximate integration: trapezium and simpson's rule)

PHY 101: GENERAL PHYSICS I (3 Units) (MECHANICS, THERMAL PHYSICS AND WAVES)

Space and Time, Units and dimensions, Kinematics; Fundamental laws of mechanics, static and dynamics, work and energy; conservation laws. Moments and energy of rotation; simple harmonic motion, motion of simple systems; Elasticity; Hooke's law, Young's shear and bulk moduli, Hydrostatics; pressure, buoyance, Archimedes' Principles; surface tension; adhesion, cohesion. Capillarity, drops and bubbles; temperature; heat; gas laws; laws of thermodynamics; kinetic theory of gases; sound. Types and properties of waves as applied to sound and light energies. Superposition of waves. Propagation of sound in gases, solids and liquids and their properties. The unified spectra analysis of waves. Applications

PHY 107: GENERAL PRACTICAL PHYSICS I (1 Unit)

This introductory course emphasis quantitative measurement, the treatment of measurement, errors and graphical analysis. A variety of experimental techniques will be employed. The experiments include studies of meters, the oscilloscope mechanical systems, mechanical, electrical resonant systems, light, heat, viscosity etc. covered in PHY101 and PHY 102. However, emphasis should be placed on the basis physical techniques for observation, measurements, data collection, analysis and deduction.

SEG 102: PRINCIPLES OF PROGRAMMING I (3 Units)

Introduces the basic principles of programming and the fundamentals of procedural and object-oriented language. Emphasis should be placed on such principles as procedural and data

abstraction, encapsulation, code reuse and composition. Ample programming labs and projects form part of this course.

SEG 104: INTRODUCTION TO WEB TECHNOLOGIES (2 Units)

Broadly examines the key technologies and programming models of the web and its underlying Internet infrastructure including client-side technologies such as HTML, CSS, and Javascript, content formats and server-side technologies such as web and application servers, database backends, client-server programming model, communication protocols such as http and tcp/ip. Students should also be introduced to content creation with client-side technologies and time permitting, some basic database-driven application using a familiar application framework.

GST 102: COMMUNICATION IN ENGLISH II (2 Units)

The course is designed to address the areas of speaking, reading, comprehension and communication. There are two significant categories of topics, sound, and literary appreciation. Students would be required to read and analyse texts from the genres of literature.

GST 106: LOGIC, PHILOSOPHY AND HUMAN EXISTENCE (2 Units)

This course shall introduce students to the understanding of scientific and critical reasoning. The study includes the broad discussions around Philosophy and its branches. With the application of Systematization, including inductive and deductive models. It also covers the spectrum of the history of human existence.

GST 110: CONTEMPORARY HEALTH ISSUES (2 Units)

The purpose of this course is to introduce students to concepts of health and wellness. Diet, exercise, diseases, drugs and substance use are all issues affecting human lives. This course presents an overview of these issues and provides basic knowledge on wellness and health as concepts. Students will also familiarize themselves with current illnesses and associated issues.

GST 111: INTRODUCTION TO MUSIC (0 Unit)

To bring to fruition the university brief in relation to music as a critical aspect of culture. The course affords students to learn basic theories in music. The course also affords students to acquire basic skills in a musical instrument of their choice.

MTH 102: ELEMENTARY MATHEMATICS II (Calculus) (3 Units)

Functions of a real variable, graphs, limits and idea of continuity. The derivative, as limit of rate of change. Techniques of differentiation, maxima and minima. Extreme curve sketching, integration, definite integrals, reduction formulae, application to areas, volumes (including approximate integration: trapezium and Simpson's rule)

PHY 102: GENERAL PHYSICS II (3 Units)

(Electricity, Magnetism and Modern Physics)

Electrostatics; conductors and currents; dielectrics; magnetic fields and electro-magnetic induction; perform's equations; electromagnetic oscillations and waves; Coulomb's law; methods of charging; ohm's law and analysis of DC circuits; AC voltages applied to Inductors, capacitors and resistance; Applications.

PHY 108: GENERAL PRACTICAL PHYSICS II (1 Unit)

This is a continuation of the experiments designed for PHY 101 and PHY 102. Some of which have been covered under PHY 107

CSC 201: COMPUTER PROGRAMMING I (3 Units)
Introduction to problem solving methods and algorithm development, designing, coding, debugging and documenting programmes using techniques of good programming language style, programming language and algorithm development. A widely used programming language should be used in teaching the above. E.g. C programming.

CSC 203: FOUNDATIONS OF SEQUENTIAL PROGRAMMING (3 Units)
The relationships between H/L languages and the architecture that underlies their implementation: basic machine architecture, assembles specification and translation of P/L Block Structured Languages, parameter passing mechanisms.

CSC 204: FUNDAMENTALS OF DATA STRUCTURES (3 Units)
Primitive types, Arrays, Records Strings and String processing, Data representation in memory, Stack and Heap allocation, Queues, TREES: Implementation Strategies for stack, queues and trees. Run time Storage management; Pointers and References, linked structures.

SEG 207: PRINCIPLES OF OPERATING SYSTEMS (3 Units)
Survey methods and algorithms used in operating systems. Concurrent distributed operation is emphasised. The main topics covered are an introduction to operating systems, process management, process schedulling, inter-process communications, memory management techniques, virtual memory, I/O management, deadlock avoidance, file system design, socket programming, distributed operation, distributed data, performance evaluation, protection and security.

CSC 208: DISCRETE STRUCTURE (3 Units)
Basic Set Theory: Basic definitions, Relations, Equivalence Relations Partition, Ordered Sets. Boolean Algebra & Lattices, Logic, Graph theory: Directed and Undirected graphs, Graph Isomorphism, Basic Graph Theorems, Matrices; Integer and Real matrices, Boolean Matrices, Matrices med m, Path matrices. Adjacency Vectors/Matrices: Path adjacency matrix, Numerical & Boolean Adjacency matrices. Applications to counting, Discrete Probability Generating Functions. Others topics covered in this course include functions, relations, propositional and first order predicate logic, proofs, and their construction, counting and elementary probability.

SEG 201: INTRODUCTION TO SOFTWARE ENGINEERING (3 Units)
Software Engineering concepts; design, development and testing of software systems. Introduction to software life cycle: requirements analysis, software architecture and design, implementation, integration, test planning, and maintenance. Review of principles of object-orientation. Object-oriented analysis using UML. Frameworks and APIs. Introduction to the client-server architecture. Analysis, design and programming of a team project including user-interface considerations.

SEG 202: SOFTWARE CONSTRUCTION (2 Units)
Topics include specifications, abstraction techniques including typing, access control,

inheritance, polymorphism, genericity and design pattern, frameworks, and architectures. Students will also learn the proper engineering use of techniques such as information hiding, classes, objects, inheritance, design by contract, exception handling, event-based systems, and concurrency. Special emphasis should be placed on suitable and appropriate object-oriented software development such as software quality and corresponding concepts, principles and best practices for addressing both functional and non-functional requirements of the software system in its architecture.

SEG 203: WEB APPLICATION DEVELOPMENT (3 Units)

This course covers client-server model for web applications and associated client-side and server side technologies, MVC development guideline and development of a complete web application using a framework such as Ruby on Rails or Django.

SEG 204: PRINCIPLES OF PROGRAMMING II (3 Units)

Builds on SEG 102 advanced concepts of object-oriented programming, language implementation models, and development with a suitable object-oriented language. Topics include input/output, networking, threading and GUI programming. Ample programming labs and projects form part of this course.

SEG 205: SOFTWARE REQUIREMENTS AND DESIGN (2 Units)

Definition of a software requirement, product and process requirements, functional and non-functional requirements, emergent properties, quantifiable requirements, and system and software requirements; Requirement process- process models and actors, process support and management, and process quality and improvement; Requirements elicitation, requirement sources, and elicitation techniques.

SEG 206: LOGIC AND ITS APPLICATIONS IN COMPUTER SCIENCE (3 Units)

Topics include: Valid & Invalid arguments; translating from English to the language of propositional and predicate logic; formal deduction and its role in proving the validity of an argument; logic & computer science- how to build circuits from logic gates and how to minimise circuits using proportional logic; introduction to Prolog- a programming language based on logic; and the applications of logic in computer science- AI, automated theorem-provers, expert systems, and so on.

SEG 208: SOFTWARE ENGINEERING PROCESS (2 Units)

Software process definition – software process management and infrastructure, Software life cycles – categories of software processes, software life cycle models, software process adaptation, practical considerations; Software process assessment and improvement - software process assessment methods, software process improvement models, and continuous and staged software process rating; Software measurement – software process and product measurement, quality of measurement results, and software process measurement techniques; Software Engineering process tools.

MTH 205: LINEAR ALGEBRA (2 Units)

Vector space over the real field. Subspaces, linear independence, basis and dimension. Linear transformations and their representation by matrices- range, null space, rank. Singular and non-singular transformation and matrices. Algebra of matrices.

GST 205: ENTREPRENEURSHIP STUDIES I (2 Units)

Introductory Entrepreneurial Skills: Relevant Concepts: Enterprise, Entrepreneur, Entrepreneurship, Business, Innovation, Creativity, Enterprising and Entrepreneurial Attitude and Behaviour. History of Entrepreneurship in Nigeria. Rationale for Entrepreneurship, Creativity and Innovation for Entrepreneurs. Leadership and Entrepreneurial Skills for coping with challenge. Unit Operations and Time Management. Creativity and Innovation for Entrepreneurship. Forms of Businesses, Staffing, Marketing and the New Enterprise. Feasibility Studies and Starting a New Business. Determining Capital Requirement and Raising Capital. Financial Planning and Management. Legal Issues, Insurance and Environmental Considerations.

GST 206: ENTREPRENEURSHIP STUDIES II (2 Units)

Profiles of business ventures in the various business ventures such as: Soap/Detergent, Tooth brush and Tooth paste making; Photography; Brick making; Rope making; Brewing; Glassware production/Ceramic production, Paper production; Water treatment/conditioning/packaging; Food processing/preservation/packaging; Metal fabrication; Tanning industry; Vegetable oil extraction; Farming; Fisheries/Aquaculture; Plastic making; Refrigeration/air-conditioning; Carving; Weaving; Bakery; Tailoring; Printing; Carpentry, Interior Decoration; Animal husbandry, etc. Case study methodology applied to the development and administration of cases that bring out key issues of business environment, start-up, pains, and gains of growth of businesses, etc, with particular reference to Nigerian businesses. Experience sharing by business actors in the economy with students during case presentations

GST 207: NIGERIA'S PEOPLE, CULTURE, GOVERNMENT AND ECONOMY(2 Units)

The course shall introduce students to the background and evolution of Nigerian State. The course shall also give students the basic understanding of the socio-cultural values of Nigeria, in terms of governance, economic, and socio-cultural values.

The course will include lectures cum research and term paper discussion, which are assigned to students: Oral and written presentation of assignments, library research, teamwork and small group presentations, and other activities. Students are required to come to class, to participate in scheduled activities. They will also be encouraged to share experiences, knowledge, talents, and skills, which are related to the entire philosophy of the course.

GST 208: COMMUNICATION IN FRENCH I (1 Unit)

The communication objectives of the course include getting acquainted with somebody: Introducing oneself, Apologizing, Saying one's name, Asking someone for her/his name. The tasks to be realised in the course amongst others pertain to meeting and greeting people, creating a contact list for the class, discovering French alphabet, identifying some objects using politeness formula.

GST 211: INSTRUMENTATION IN MUSIC (0 Unit)

Students will learn how to play an instrument of their choice including voice

CSC 301: STRUCTURED PROGRAMMING (3 Units)

Structured Programming elements, structured design principles, abstraction modularity, stepwise

refinement, structured design techniques. Teaching of a structured programming language (e.g. C++, Python).

CSC 309: SYSTEMS ANALYSIS AND DESIGN: (3 Units)

System Concept, organization of a data processing department, Feasibility study: project identification and selection fact-finding and analysis; Process of System Design, design problem identification, definitions and solutions, physical and implementation, data capture, data recording transmission, conversion and possible effect, file design control and security. Personnel training, system testing and maintenance, evaluation process, system documentation, reporty writing and presentation. System Development Life Cycle Analysis: Fact gathering Techniques, data flow diagrams, Process description data modeling: Structure Charts, form designs, security, automated Tools for design.

CSC 313 OBJECT-ORIENTED PROGRAMMING: (3 Units)

Basic OOP Concepts: Classes, Objects, inheritance, polymorphism, Data Abstraction, Tools for developing, Compiling, interpreting and debugging, Java Programs, Java Syntax and data objects, operators. Central flow constructs, objects and classes programming, Arrays, methods. Exceptions, Applets and the Abstract, OLE, Persistence, Window Toolkit, Laboratory exercises in an OOP Language. **Pre-requisite: CSC 201 or 202.**

SEG 301: FORMAL METHODS IN SOFTWARE DEVELOPMENT (2 Units)

Topics from process improvement: software configuration management, Formal specification, software cost estimate, software Architecture, Software patterns, Software Reuse and Open source development.

SEG 303 – MODEL-BASED SOFTWARE ENGINEERING (2 Units)

The role and nature of model-based methods in Software Engineering; The theory of modelling language design; UML modelling language; Approaches to model transformations and automated code generation; Model analyses methods; Basics of model-based software engineering tooling; Key elements of model-based Software Engineering methodologies; Industrial application of model-based Software Engineering (experience and issues); System and multi-model development; Architectural design using model-based software engineering; Relevant research areas and directions; model-based Software Engineering process models (e.g., agile modelling).

SEG 305: SOFTWARE TESTING & QUALITY ASSURANCE (2 Units)

How to assure it and verify it, and the need for a culture of quality. Avoidance of errors and other quality problems. Inspections and reviews. Testing, verification and validation techniques. Testing-related terminology, key issues, and relationships of testing to other activities (testing versus static software quality management techniques, testing versus correctness proofs and formal verification, testing versus debugging, testing versus programming); Test levels – the target of the test, and objectives of testing; Test techniques – software engineer's intuition and experience, input domain-based techniques, code-based techniques, fault-based techniques, usage-based techniques, model-based techniques, techniques based on nature of application, and selecting and combining techniques; Test-related measures - evaluation of the program under test, and evaluation of the tests performed; Test process - practical considerations, and test activities; Software testing tools – testing tool support and

categories of tools. Process assurance vs. Product assurance. Quality process standards. Statistical approaches to quality control.

SEG 307: SOFTWARE ENGINEERING II (3 Units)

Introduction to Software Engineering, SDLC: Understand the need of Software Engineering and its origins, familiarising the student with the challenges, code of conduct and ethics of Software Engineering, and get acquainted with the basic concepts of SDLC, Software Models, Agile Methodologies. Compare the traditional waterfall model to the incremental model, the agile model and other appropriate models as well; Software Requirements Engineering and Architecture; USE Case Modelling and UML; Analysis and Design: Identifying Concepts, Concept Associations and Attributes, Domain Analysis, Contracts: Preconditions and Postconditions, Design: Assigning Responsibilities, Design Principles for Assigning Responsibilities, Class Diagram, Why Software Engineering Is Difficult; Implementation, Testing; Introduction to Modelling and System Specifications: Overview of Software Testing, Test Coverage and Code Coverage, Practical Aspects of Unit Testing, Integration and Security Testing, Test-driven Implementation, Refactoring; Software Measurement and Estimation; Software Reengineering and Documentation.

SEG 309: COMPUTER ARCHITECTURE & ORGANISATION (3 Units)

Differences between computer architecture and computer organisation, basic structure of computers; performance evaluation: metrics and calculations, performance equations, Amdahl's law; CPU organisation and micro-architectural level design; instruction set design; register transfer; RISC design principles; data-path design; controller design; memory system; addressing; microprogramming; computer arithmetic; survey of real computers and microprocessors; peripheral devices and input/output buses; and introduction to parallel computing. The course is a broad introduction to all aspects of computer systems organisation and architecture and serves as the foundation for subsequent computer systems courses.

SEG 311: DESIGN AND ANALYSIS OF COMPUTER ALGORITHMS (3 Units)

Introduction to Algorithm and its importance, mathematical foundations; growth functions, complexity analysis of algorithms, summations, recurrences, sorting algorithms. Algorithm design: divide-and-conquer approach, greedy approach. Graph algorithms: graph searching, topological sort, minimum spanning tree, shortest paths, backtracking and its applications in games. String matching. Dynamic programming and longest common subsequence. Theory of NP-completeness. Turing machines and the halting problem.

SEG 313: DATA STRUCTURES AND ALGORITHMS (3 Units)

The course covers Abstract Data Types (ADTs) and their support and implementations in object-oriented languages. Topics include recursion, complexity analysis, linear data structures (stacks, queues, priority queues, lists and strings), and non-linear data structures (has tables, binary trees, search trees, balanced trees, heaps), searching and sorting algorithms and graph algorithms. This will also include substantial programming assignments and projects. Introduction to algorithm for parallel and distributed computing.

SEG 315: OBJECT-ORIENTED ANALYSIS AND DESIGN (3 Units)

Object-oriented approach to information system development, particularly in reference to the earlier stages of analysis and design. Importance of modelling, principles of modelling, object-

oriented modeling, conceptual model of the Unified Modelling Language (UML), architecture, software development life cycle. The principles and basic concepts of object orientation and the different aspects of object-oriented modeling as represented by the UML technique. Case study of a typical UML-based CASE tool.

SEG 399 INDUSTRIAL TRAINING (SIWES) (16 Units)

Student's Industrial Work Experience of 6 months duration. Student's reports will be presented in a seminar.

MTH 315: OPERATIONS RESEARCH (3 Units)

Nature and Scope of Operations Research. Linear programming and graphical, simplex (including big M and two-phase) methods. Sensitivity analysis. Duality theory. Transportation and assignment problems. Network analysis: CPM and PERT. Inventory theory and applications. Sequencing and scheduling.

GST 308: COMMUNICATION IN FRENCH II (1 Unit)

The course builds on the foundations already laid in GST 208 by expatiating on the communication objectives of getting acquainted with somebody: Introducing oneself, Apologizing, Saying one's name, Asking someone for her/his name. The tasks to be realised in the course amongst others pertain to meeting and greeting people, creating a contact list for the class, discovering French alphabet, identifying some objects using politeness formula.

ENT 301: ENTREPRENEURSHIP STUDIES III (2 units)

The course equips students with the tools and capacity needed to start and grow a successful business venture, to identify and maximise opportunities, and to engage in product development that meets target market needs.

ENT 302: ENTREPRENEURSHIP SKILLS (2 units)

This is a practical entrepreneurship course. In this two course students are exposed to the ***"Chrisland University Sustainability Project (CLUSP)"*** which has a focus of connecting entrepreneurship teaching and theory into student generated sustainable development products and ideas that solve critical real life problems.

LDS 401: LEADERSHIP STUDIES (1 Unit)

The course describes leadership, leadership traits, leadership skills, theories of leadership, leadership and self-awareness, the role of identity in leadership, vision in leadership, the changing nature of leadership, leading the organisation, leadership effectiveness, and managing change in organisations.

CSC 406: COMPUTER GRAPHICS AND VISUALIZATION (2 Units)

Hardware aspect, plotters microfilm, plotters display, graphic tablets, light pens, other graphical input aids Facsimile and its problems, refresh display, refresh huggers, changing images, light pen interaction. Two and three dimensional transformation, perspective of clipping algorithms.

Hidden line removal and bolded surface removal. Warmock's method; shading, data reduction for graphical input. Introduction to CAD writing and character recognition; Curve synthesis and fitting, Contouring, Ring structures versus doubly linked lists. Hierarchical structures, Data structure, Organization for interceptive graphics.

CSC 408: MODELLING AND SIMULATIONS (3 Units)

Basic Definitions and Uses, Simulation Process, Some basic statistic Distributions; Theory, Model and Simulation. Queues; Basic components, Kendal notation, Queuing rules, Little's Law, Queuing networks, Special/types of queues. Stochastic Processes; Discrete state and continuous state processes, Markov processes, Birth-Death Processes, Poisson Processes. Random Numbers; types of Random, Number Exercises.

CSC 411: COMPUTER NETWORKS AND COMMUNICATION (2 Units)

Introduction; wares, Fourier analysis, measure of communication, channel characteristics, transmission media, noise and distortion, modulation and demodulation, multiplexing (TDM, FDM and FCM), Parallel and serial transmission (synchronous vs. Asynchronous). Bus structures and loop systems, computer network Examples and design consideration, data switching principles, broadcast techniques, network structure for packet switching, protocols, description of network e.g. ARPANET, etc.

SEG 401: SOFTWARE CONFIGURATION MANAGEMENT & MAINTENANCE (2 Units)

Management of the software configuration process- organisation context for software configuration management, constraints and guidance for software configuration management process, planning for software configuration management, and surveillance of software configuration management; Software configuration identification and software library; Software configuration control requesting, evaluating and approving software changes, implementing software changes, and deviations and waivers; software configuration status accounting; software configuration status information and reporting; software configuration auditing; key issues in software maintenance- technical issues, management issues, maintenance cost estimation, and software maintenance measurement; maintenance process- maintenance processes and activities; techniques for maintenance- program comprehension, reengineering, reverse engineering, migration, and retirement.

SEG 402: ENGINEERING MOBILE APPLICATIONS (2 Units)

Introduction to developing mobile applications, beginning with mobile operating systems capabilities and application architecture and extending to major components, such as activities, services, broadcast receivers, etc. Development of interactive applications using widget libraries, web-based services, animation, an SQL database engine, and multithreading.

SEG 403: SOFTWARE ENGINEERING PROFESSIONAL PRACTICE (2 Units)

Accreditation, certification and licensing, codes of ethics and professional conduct, nature and role of professional societies and Software Engineering standards, economic impact of software, employment contracts, legal issues, documentation, and trade-off analysis; Group dynamics and psychology – dynamics of working in teams/groups, individual cognition, dealing with problem complexity, interacting with stakeholders, dealing with uncertainty and ambiguity, and dealing with multicultural environments; Communication skills – reading,

understanding and summarising, writing, teams and group communication, and presentation skills.

SEG 404: SOFTWARE ENGINEERING PROJECT MANAGEMENT (2 Units)
Determination and negotiation of requirements, feasibility analysis, and process for the review and revision of requirements; Software project planning – process planning, determine deliverables, effort, schedule and cost estimation, resource allocation, risk management, quality management, and plan management; Software project enactment – implementation of plans, software acquisition and supplier contract management, implementation of measurement process, monitor process, control process, and reporting; Review and evaluation – determining satisfaction of requirements, and reviewing and evaluating performance; Closure – determining closure and closure activities; Software engineering measurement – establish and sustain measurement commitment, plan the measurement process, perform the measurement process; and evaluate measurement; Software Engineering management tools.

SEG 405: CONCEPTS OF PROGRAMMING LANGUAGES (3 Units)
Preliminaries, evolution of programming languages, paradigms, language design considerations, language processing including syntax and semantic analysis, naming, binding, type checking, expression, and assignment statement, statement-level control structures, subprograms, abstract data types, support for object-oriented languages, concurrency, exception handling, functional and logic programming.

SEG 406: SOFTWARE ENGINEERING ECONOMICS (2 Units)
Software Engineering economics fundamentals; lifecycle economics; Risk and uncertainty - goals, estimates and plans, estimation techniques, addressing uncertainty, prioritization, decisions under risk and uncertainty; Economic analysis methods – for-profit decision analysis, minimum acceptable rate of return, return on investment and capital employed, cost-benefit analysis, cost-effectiveness analysis, break-even analysis, business case, multiple attribute evaluation, and optimization analysis; Practical considerations – the "good enough" principle, friction - free economy, ecosystems, and offshoring and outsourcing.

SEG 407: RESEARCH METHODOLOGY (1 Unit)
Foundations of research; problem identification and formulation; research design; qualitative and quantitative research; measurement; sampling; data analysis; interpretation of data and paper writing; use of encyclopaedias, research guides, handbook, etc., academic databases for computer science discipline; use of tools/techniques for research; reference management software, software for detection of plagiarism.

SEG 408: SOFTWARE ENGINEERING MODELS AND METHODS (2 Units)
Modelling principles, properties and expression of models, syntax, semantic and pragmatics, and preconditions, post-conditions, and invariants; Analysis of models – analyzing for completeness, consistency and correctness, traceability and interaction analysis; Software engineering methods – heuristic methods, formal methods, prototyping methods and agile methods.

SEG 409: HUMAN-COMPUTER INTERACTION (3 Units)

Introduces the principles of user interface development, focusing on three key areas: 1. Design: How to design user interface, starting with human capabilities and using those capabilities to drive design techniques: task analysis, user-centred design, iterative design, usability guidelines, interaction styles, and graphic design principles. 2. Implementation: Techniques for building user interfaces, including low-fidelity prototypes, Wizard of Oz, and other prototyping tools; input models, output models, model-view-controller, layout, constraints, and toolkits. 3. Evaluation: Techniques for evaluating and measuring interface usability, including heuristic evaluation, predictive evaluation, and user testing.

SEG 412: OPEN SOURCE DEVELOPMENT AND APPLICATION (2 Units)

The course introduces concepts, principles, and applications of open source software. Discusses about open source software development process. Covers economy, business, societal and intellectual property aspects of open source software. Entails hands-on experiments on open source software and related tools through developing various open source software applications such as mobile applications and web applications building on existing open-source frameworks and application development platforms.

SEG 413: DATABASE SYSTEMS (3 Units)

Concepts, principles, components, development and applications of database systems. The conceptual models and structures necessary for designing and implementing a relational database system. Entity-Relationship, Relational Data Models, Relational Algebra, SQL, Normalisation, File Organisation, Indexing, Hashing, and Enterprise-wide web-based applications that employ database.

SEG 414: DISTRIBUTED, PARALLEL AND CLOUD COMPUTING SYSTEM (2 Units)

Analysis and Design of Parallel and Distributed Algorithms; Languages/Operating Systems for parallel processing; GPGPU computing; Architecture of parallel/distributed systems, Tools for parallel computing; Parallel (distributed) database systems; Networking aspects of parallel/distributed computing; Parallel/distributed scientific computing applications; High-performance computing applications in molecular sciences; multimedia applications for parallel/distributed systems; Grid networks; services and applications; Distributed File Systems; Hyper Scale/Hyper Converged Distributed Storage Design, Storage I/O Protocols; Cloud as a Service; Cloud Infrastructure, Management and operations, Performance, Scalability, Reliability, Virtualisation, Loud Provisioning Orchestration, Architecture support, Development tools, Platforms, and Applications. Legal aspects and Service Level Agreement, Mobile computing advances in the cloud, Performance optimisation.

SEG 416: SOFTWARE ENGINEERING SECURITY (2 Units)

History and terminology, security mindset, design principles, system/security life cycle, security implementation mechanisms, information assurance analysis model, disaster recovery, and forensics; security mechanisms- cryptography, authentication, redundancy, and intrusion detection; operational issues-trends, auditing, cost/benefit analysis, asset management, standards, enforcement, legal issues, and disaster recovery; policy- creation of policies, maintenance of polices, prevention, avoidance, incident response (forensics), and domain integration (physical, network, internet, etc.); Attacks- social engineering, denial of service, protocol attacks, active and passive attacks, buffer overflow attacks, and malware; security domains- security awareness and possible domains; forensics legal systems, digital forensics and its relationship to other

forensic disciplines, rules of evidence, search and seizure, digital evidence, and media analysis; security services; threat analysis model; vulnerabilities.

SEG 418: SOFTWARE ARCHITECTURE AND DESIGN (2 Units)

An in-depth look at software design. Continuation of the study of patterns, frameworks, and architectures. Survey of current middleware architectures. Design of distributed systems using middleware. Component-based design. Measurement theory and appropriate use of metrics in design. Designing for qualities such as reliability, performance, safety, security, reusability, etc. Measuring internal qualities and complexity of software. Evaluation and evolution of designs. Basics of software evolution, reengineering, and reverse engineering.

SEG420: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS (2 Units)

The evolution of computing, defining artificial intelligence, general problem solving approaches in artificial intelligence., characteristic requirements for the realisation of intelligent systems, programming languages for artificial intelligence and architecture for artificial intelligence machines. The psychological perspective of cognition; production systems; problem solving by intelligent search; the logic proposition and predicates; default and non-monotonic reasoning. Structural approach to knowledge representation, the nature and goals of soft computing. The nature of expert systems; theoretical foundations; basic forms of inference; the representation and manipulation of knowledge in a computer; rule-based representations; logic-based representations; frames; semantic and partitioned nets; basic components of an expert system; generation of explanations; handling of uncertainties; truth maintenance systems; expert system architectures; an analysis of some classic expert systems; building expert systems; methodologies for building expert systems; knowledge acquisition and elicitation, formalisation, representation and evaluation; knowledge engineering tools; expert systems paradigms.

SEG 422: SPECIAL TOPICS IN SOFTWARE ENGINEERING (2 Units)

Recent topics and developments in software engineering are expected to be introduced from year to year. Apart from seminars to be delivered by lecturers or guests, students are expected to do substantial readings on their own.

SEG 424: FAULT TOLERANT COMPUTING (2 Units)

Introduction and overview of fault tolerant schemes; fault and error modelling; test generation and fault simulation; concepts in fault-tolerance; reliability/availability modelling; system level diagnosis; low level fault-tolerance- coding techniques (basic principles, parity hit codes, hamming codes, error detection and retransmission codes, burst error correction codes, Reed-Solomon codes, etc.); high-level fault tolerant techniques in systems; rollback, check pointing, reconfiguration; software fault-tolerance; fault tolerant routing; integrated hardware/software fault-tolerance; redundancy, spares and repairs- apportionment, systems redundancy; software reliability and recovery techniques, network system reliability, reliability optimisation.

SEG 426: EMBEDDED SYSTEMS (2 Units)

Introduction to embedded computing and embedded systems; typical embedded systems- core of the embedded system, memory, communication interface, embedded firmware; embedded real-time operating systems; real-time operating systems-based embedded system design; task communications and synchronisation.

SEG 428: GAME DESIGN AND DEVELOPMENT (2 Units)

The course covers game development history, platforms, goals and genres, player elements, story and character development, gameplay, levels, interface, audio, development team roles, game development process, and marketing and maintenance. Students will play games, analyse them, and complete portions of game designs with appropriate documentation.

SEG 430: MODELLING AND COMPUTER SIMULATION (2 Units)

Introduction to simulation concepts, introduction to models, problem formulation, project planning, system definition, input data collection and analysis, modelling translation, verification, validation, experimental design, analysis, project reports and presentations, training simulators.

SEG 499: FINAL YEAR PROJECT (6 Units)

An independent or group investigation of appropriate software, hardware, communication and networks or IT related problems in Software Engineering carried out under the supervision of a lecturer. Before registering, the student must submit a written proposal to the supervisor to review. The proposal should give a brief outline of the project, estimated schedule of completion, and computer resources needed. A formal written report is essential and an oral presentation may also be required.

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CCMAS

100 LEVEL FIRST SEMESTER

Course Code	Course Title	Units	Status	LH	PH
MTH 101	Elementary Mathematics I	2	C	30	0
PHY 101	General Physics I	2	C	30	0
PHY 107	General Practical Physics I	1	C	0	45
STA 111	Descriptive Statistics	3	C	45	0
COS 101	Introduction to Computing Sciences	3	C	30	45
CLU-CSC 103	Computer Hardware and Maintenance	3	C	15	45
GST 101	Communication in English I	2	C	30	0
GST 107	Use of Library, Study Skills and ICT	2	C	30	0
GST 111	Introduction to Music	0	C	15	0
	Total	**18**			

100 LEVEL SECOND SEMESTER

Course Code	Course Title	Units	Status	LH	PH
CLU-SEG 104	Fundamentals of Cloud Computing	3	C	45	15
CLU-CSC 134	Web Programming 1	3	C	15	45
MTH 102	Elementary Mathematics II	2	C	30	0
PHY 102	General Physics II	2	C	30	0
PHY 108	General Practical Physics II	1	C	0	45
COS 102	Introduction to Problem Solving	3	C	30	45
GST 102	Communication in English II	2	C	30	0
GST 106	Logic, Philosophy and Human Existence	2	C	30	0
GST 110	Contemporary Health Issues	2	C	30	0
	Total	**20**			

200 LEVEL FIRST SEMESTER

Course Code	Course Title	Units	Status	LH	PH
CLU-SEG 203	Block-Chain Technology	3	C	15	45
CLU-SEG 231	Introduction to Green Computing	3	E	45	15
MTH 201	Mathematical Methods I	2	C	30	0
COS 201	Computer Programming I	3	C	30	45
SEN 201	Introduction to Software Engineering	2	C	30	0
CSC 203	Discrete Structures	2	C	30	0
IFT 211	Digital Logic Design	2	C	15	45
GST 205	Entrepreneurship Studies I	2	C	30	30
GST 207	Nigeria's People, Culture, Government and Economy	2	C	30	0
GST 209	Morality, Religion, Ethics and Etiquette	1	C	15	0
GST 211	Instrumentation in Music	0	C	15	0
	Total	**22**			

200 LEVEL SECOND SEMESTER

Course Code	Course Title	Units	Status	LH	PH
CLU-SEG 208	System and Software Specification and Requirements	3	C	15	45
CLU-SEG 222	Software Engineering	3	C	30	30
MTH 202	Mathematical Methods II	2	C	30	0
COS 202	Computer Programming II	3	C	30	45
INS 204	System Analysis and Design	3	C	30	45
IFT 212	Computer Architecture and Organisation	2	C	15	45
GST 206	Entrepreneurship Studies II	2	C	30	30
GST 204	Peace and Conflict Resolution	1	E	30	0
GST 208	Communication in French	1	C	15	0
	Total	**20**			

300 LEVEL FIRST SEMESTER

Course Code	Course Title	Units	Status	LH	PH
CLU-SEG 311	Software Project Management I	3	C	15	45
CLU-SEG 331	Software Defect Management	3	C	15	45
CLU-SEG 323	Foundation to Machine Learning	3	C	30	30
CLU-SEG 325	Cryptography and Computer Security	3	C	15	45
SEN 301	Object-Oriented Analysis and Design	2	C	15	45
CSC 301	Data Structures	3	C	15	45
ENT 301	Entrepreneurship Studies III	2	C	30	30
	Total	**19**			

300 LEVEL FIRST SEMESTER

Course Code	Course Title	Units	Status	LH	PH
GST 312	Peace and Conflict Resolution	2	C	30	0
ENT 312	Venture Creation	2	C	15	45
SEN 304	Software Testing and Quality Assurance	2	C	15	45
SEN 306	Software Construction	2	C	15	45
SEN 322	Software Engineering Innovation and New Technology	2	C	15	45
SEN 399	SIWES II	3	C	0	135
CSC 308	Operating Systems	3	C	30	45
ENT 302	Entrepreneurship Skills	2	C	30	30
GST 308	Communication in French II	1	C	15	0
	Total	**19**			

400 LEVEL FIRST SEMESTER

Course Code	Course Title	Units	Status	LH	PH
CLU-SEG 411	Software Engineering 2	3	E	45	15
CLU-SEG 413	Cloud Computing and Internet of Things	3	C	15	45
CLU-SEG 415	Computer Aided Software Engineering	3	C	15	45
COS 409	Research Methodology and Technical Report Writing	3	C	30	0
SEN 401	Software Configuration Management and Maintenance	2	C	15	45
INS 401	Project Management	2	C	30	0
SEN 497	Final Year Student's Project I	3	C	0	135
LDS 401	Leadership Studies	1	C	15	0
	Total	**20**			

400 LEVEL SECOND SEMESTER

Course Code	Course Title	Units	Status	LH	PH
CLU-SEG 408	Advanced Software Testing	3	C	15	45
SEN 410	Software Architecture and Design	2	C	15	45
SEN 498	Final Year Student's Project II	3	C	0	135
	Total	**8**			

Electives: Student must register for any combinations of the following elective courses to make a minimum of 15units:

Course Code	Course Title	Units	Status	LH	PH
CLU-CSC 434	Deep Learning	3	C	30	30
CSC 402	Ethics and Legal Issues in Computer Science	2	C	30	0
CLU-CYB 412	Big Data	3	E	45	15
CLU-CYB 430	Fault-Tolerant Computing	3	E	15	45
CYB 404	Cloud Computing Security	2	E	30	0

COURSE CONTENTS AND LEARNING OUTCOMES

CLU-CSC 103 Computer Hardware and Maintenance (3 Units: (Compulsory): 15H (Lecture) 45H (Practical)

Senate Approved Relevance

Training of graduates who are highly skilled in not only software development but also in hardware maintenance. The knowledge will inculcate in the students technical know-how and on-site experience that will guide them in recommending, selecting, and procuring state of art devices either for private and public use. Also, the course will equips the students on steps to follow to maintain organizational information technological infrastructure and therefore shine the light forth as contained in the vision and mission statement of the university as embedded in the motto of the school "shine forth the Light". Relevance is seen in CLU desire to have personalities that take care and maintain the multi-million naira IT infrastructures put in place to enhance teaching and learning process in the university.

Overview

This course is an exploratory, first course in computer hardware systems and maintenance designed primarily for students in computer science. However, it also meets the need of students in other fields, as a course that provides hands-on training in the installation, configuration, optimization and upgrading of computer systems. As a practical course, the focus is to impart useful skills on the students in order to enhance ability to install, troubleshoot, repair and maintain not only computer systems but also computer laboratories.

Objectives:

The objectives of the course are to:

1. Explain the computer system;
2. Explain computer components;
3. Identify hardware components;
4. Explain computer system configuration;
5. Disassemble and assemble a computer system
6. Explain types of maintenance;
7. Use basic linux and windows operating commands;
8. Use common diagnostic tools;
9. Measure circuit voltage of a component;
10. Carry-out back up procedure;
11. Identify computer network devices;
12. Junks removal;
13. Set up small local area network.

Learning outcomes

On completion of the course, students should be able to:

1. List at least five components of computer system;
2. State at least 3 different processor configurations;
3. Disassemble and assemble a computer system;
4. List 2 types of maintenance;
5. List at least 6 basic linux/windows operating system troubleshooting and maintenance commands;
6. State at least 10 diagnostic software tools;
7. State at least 4 diagnostic hardware tools;
8. Identify a short circuit component;
9. Back up procedure both online and offline;
10. Mention at least 4 computer network devices;
11. Set up small local area network consist of at least 3 nodes.

Course Contents

Fundamentals of computer systems. Introduction to computer hardware. Computer system parts. Components of computer system. Hardware Identification. Configuration of computer system, Optimization and upgrading of computer systems. Computer system disassembling and assembling. Computer hardware installation. Operating system installation. Types of processor and their specification. Maintenance techniques. Types of maintenance. Computer troubleshooting and maintenance tools. Computer system troubleshooting. Computer system repair and maintenance. Diagnostic techniques. Troubleshooting and repair of computer systems. Junks removal. Computer accessories. Portable computers. Basic window and Linux command. Measurement of circuit voltage and current using multi-meter. Types of computer memory. Data back-up. Introduction to networking devices. Basic network device configuration.

Minimum Academic Standards

CCMAS Benchmarks: A laboratory/studio equipped with computing resources and tools.

MTH 101: Elementary Mathematics I (Algebra and Trigonometry) (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:

1. explain basic definition of Set, Subset, Union, Intersection, Complements and use of Venn diagrams;
2. solve quadratic equations;
3. solve trigonometric functions;
4. identify the various types of numbers; and
5. solve some problems using Binomial theorem.

Course Contents

Elementary set theory, subsets, union, intersection, complements, Venn diagrams. Real numbers; integers, rational and irrational numbers, mathematical induction, real sequences and series, theory of quadratic equations, binomial theorem. Complex numbers; algebra of complex numbers; the Argand diagram. De-Moivre's theorem, nth roots of unity. Circular measure, trigonometric functions of angles of any magnitude, addition and factor formulae.

PHY 101: General Physics I (Mechanics) (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. identify and deduce the physical quantities and their units;
2. differentiate between vectors and scalars'
3. describe and evaluate motion of systems on the basis of the fundamental laws of mechanics;
4. apply Newton's laws to describe and solve simple problems of motion;
5. evaluate work, energy, velocity, momentum, acceleration, and torque of moving or rotating objects;
6. explain and apply the principles of conservation of energy, linear and angular momentum;
7. describe the laws governing motion under gravity; and
8. explain motion under gravity and quantitatively determine behaviour of objects moving under gravity.

Course Contents
Space and time. Units and dimension, Vectors and Scalars, Differentiation of vectors. Displacement, velocity and acceleration. Kinematics. Newton laws of motion (Inertial frames, Impulse, force and action at a distance, momentum conservation). Relative motion. Application of Newtonian mechanics. Equations of motion. Conservation principles in physics, Conservative forces, conservation of linear momentum, Kinetic energy and work, Potential energy, System of particles, Centre of mass. Rotational motion. Torque, vector product, moment, rotation of coordinate axes and angular momentum. Polar coordinates. Conservation of angular momentum. Circular motion. Moments of inertia, gyroscopes and precession. Gravitation: Newton's Law of Gravitation, Kepler's laws of planetary motion, Gravitational potential energy, Escape velocity, Satellites motion and orbits.

PHY 107: General Practical Physics I (1 Unit C: PH 45)
Learning Outcomes
At the end of the course, students should be able to:
1. conduct measurements of some physical quantities;
2. make observations of events, collect and tabulate data;
3. identify and evaluate some common experimental errors;
4. plot and analyse graphs; and
5. draw conclusions from numerical and graphical analysis of data.

Course Contents
This introductory course emphasizes quantitative measurements, the treatment of measurement errors and graphical analysis. A variety of experimental techniques should be employed. The experiments include studies of meters, the oscilloscope, mechanical systems, electrical and mechanical resonant systems, light, heat, viscosity etc., covered in PHY 101 and PHY 102. However, emphasis should be placed on the basic physical techniques for observation, measurements, data collection, analysis and deduction.

STA 111: Descriptive Statistics (3 Units C: LH 45)
Learning Outcomes
At the end of the course, students should be able to:

1. explain the differences between permutation and combination;
2. explain the concept of random variables and relate it to probability and distribution functions;
3. describe the basic distribution functions; and
4. explain the concept of exploratory data analysis.

Course Contents
Permutation and combination. Concepts and principles of probability. Random variables. Probability and distribution functions. Basic distributions: Binomial, geometric, Poisson, normal and sampling distributions; exploratory data analysis.

COS 101: Introduction to Computing Sciences (3 Units C: LH 30; PH 45)
Learning Outcomes
At the end of the course, students should be able to:
1. explain basic components of computers and other computing devices;
2. describe the various applications of computers;
3. explain information processing and its roles in the society;
4. describe the Internet, its various applications and its impact;
5. explain the different areas of the computing discipline and its specializations; and
6. demonstrate practical skills on using computers and the internet.

Course Contents
Brief history of computing. Description of the basic components of a computer/computing device. Input/Output devices and peripherals. Hardware, software and human ware. Diverse and growing computer/digital applications. Information processing and its roles in society. The Internet, its applications and its impact on the world today. The different areas/programs of the computing discipline. The job specializations for computing professionals. The future of computing.

Lab Work: Practical demonstration of the basic parts of a computer. Illustration of different operating systems of different computing devices including desktops, laptops, tablets, smart boards and smart phones. Demonstration of commonly used applications such as word processors, spreadsheets, presentation software and graphics. Illustration of input and output devices including printers, scanners, projectors and smartboards. Practical demonstration of the Internet and its various applications. Illustration of browsers and search engines. How to access online resources.

GST 101: COMMUNICATION IN ENGLISH 1 (2 Units)
The course is designed to address the foundational principles and rules of grammar. Its emphasis is on grammar and usage. Thus, practical demonstrations, using appropriate examples are used for better assimilation and mastery of the rules that define how sentences are formed and used.

Learning Outcomes
At the end of this course, students should be able to:
1. identify possible sound patterns in English language;
2. list notable language skills;
3. classify word formation processes;
4. construct simple and fairly complex sentences in English;

5. apply logical and critical reasoning skills for meaningful presentations;
6. demonstrate an appreciable level of the art of public speaking and listening; and
7. write simple and technical reports.

Course Contents

Sound patterns in English Language (vowels and consonants, phonetics and phonology). English word classes (lexical and grammatical words, definitions, forms, functions, usages, collocations). Sentence in English (types: structural and functional, simple and complex). Grammar and Usage (tense, mood, modality and concord, aspects of language use in everyday life). Logical and Critical Thinking and Reasoning Methods (Logic and Syllogism, Inductive and Deductive Argument and Reasoning Methods, Analogy, Generalisation and Explanations). Ethical considerations, Copyright Rules and Infringements. Writing Activities: (Pre-writing, writing, post writing, editing and proofreading; brainstorming, outlining, paragraphing. Types of writing, Summary, Essays, Letter, Curriculum Vitae, Report writing, Note making, etc. Mechanics of writing). Comprehension Strategies: (Reading and types of Reading, Comprehension Skills, 3RsQ). Information and Communication Technology in modern language learning. Language skills for effective communication. Major word formation processes. Writing and reading comprehension strategies. Logical and critical reasoning for meaningful presentations. Art of public speaking and listening. Report writing.

GST 107: USE OF LIBRARY, STUDY SKILLS AND ICT (2 Units)

Brief history of libraries; library and education; university libraries and other types of libraries; study skills (reference services);Types of library materials, using library resources including e-learning, e-materials, etc. ; understanding library catalogue (card, OPAC, etc.) and classification; copyright and its implications; database resources; bibliographic citations and referencing. Development of modern ICT; Hardware technology; software technology; input devices; storage devices; output devices; communication and internet services; word processing skills (typing, etc.)

GST 111: INTRODUCTION TO MUSIC (0 Unit)

To bring to fruition the university brief in relation to music as a critical aspect of culture. The course affords students to learn basic theories in music. The course also affords students to acquire basic skills in a musical instrument of their choice.

CLU-SEG 104 Fundamentals of Cloud Computing (3 Units: (Compulsory): 45H (Lecture) 15H (Practical)

Senate Approved Relevance

Educating students on the fundamentals of cloud computing equips them with the much sought after knowledge of relevance in the field of computer science. It gives students the needed edge which comes with being informed about trending matters. Cloud computing is the on-demand solution for storing and retrieving data globally. It is fast emerging as an essential component for IT-based enterprises. Organizations, both big and small, have deployed cloud technology in a suitable capacity.

In line with Chrisland University's motto of cultivating intellectually radiant students, education on cloud computing avails our students timely access into a world where there is currently little mass knowledge about such and helps to produce industry ready students.

With the recent push for online courses, digital lesson plans, and remote learning, cloud-based platforms are more important than ever. The knowledge of the fundamentals of cloud computing enables student's transverse the realm of cloud computing with ease.

Overview

Cloud computing evokes different perceptions in different people. To some, it refers to accessing software and storing data in the "cloud" representation of the Internet or a network and using associated services. To others, it is seen as nothing new, but just a modernization of the time-sharing model that was widely employed in the 1960s before the advent of relatively lower-cost computing platforms.

Cloud computing however is a model for enabling convenient, on-demand network access to a shared pool of configurable and reliable computing resources (e.g., networks, servers, storage, applications, services) that can be rapidly provisioned and released with minimal consumer management effort or service provider interaction.

This course is designed cover management, storage and processing of data on networks of the internet server. This cloud computing course presents the basics of cloud computing, from application, administration to programming and infrastructure.

Objectives

The objectives of the course are to:

1. understand the various distributed system models and evolving computing paradigms.
2. gain knowledge in virtualisation of computer resources.
3. introduce the students to the fundamentals of cloud computing.
4. introduce concepts related to the analysis, design and implementation of computation and storage clouds.
5. help students understand the purpose of cloud computing.
6. support the development of knowledge about the different services offered by providers of cloud computing.
7. evaluate the security of using the cloud for storage and document/data creation

Learning outcomes

The students should be able to:

1. understand the necessary theoretical background for computing and storage clouds environments.
2. explain the levels of virtualisation and resources virtualisation

3. analyse the reasons for migrating into cloud
4. list at least five (5) concepts related to the analysis, design and implementation of computation and storage clouds.
5. explain at least three (3) different services offered by providers of cloud computing.
6. evaluate the security of using the cloud for storage and document/data creation.
7. distinguish the different models and computing paradigms.
8. to effectively use the cloud services in terms of infrastructure and operating platforms.
9. to apply the services in the cloud for real world scenarios.

Course Contents

Distributed System Models and Enabling Technologies. Scalable Computing over the Internet. System Models for Distributed and Cloud Computing. Software Environments for Distributed Systems and Clouds. Virtualisation concepts: Virtual Machines and Virtualisation of Clusters and Data Centre**.** Cloud Fundamentals**.** Private Cloud Environment. Public Cloud Environment. Auto-provisioning. Cloud Deployment Models. Types of Cloud Computing: Cloud Service Categories. Desired Features of a Cloud. Cloud Infrastructure Management. Benefits of Cloud Computing for Businesses. Cloud computing securitization. Ethics and standard of cloud. Cloud Distributed Storage. Cloud Distributed Computation. Security in Cloud. Information Security Concerns. Security Enhancing Techniques.

Minimum Academic Standards

CCMAS Benchmarks: A laboratory equipped with internet enabled capabilities. Other NUC MAS requirements.

GST 112: Nigerian Peoples and Culture (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. analyse the historical foundation of the Nigerian culture and arts in pre-colonial times;
2. list and identify the major linguistic groups in Nigeria;
3. explain the gradual evolution of Nigeria as a political unit;
4. analyse the concepts of Trade, Economic and Self-reliance status of the Nigerian peoples towards national development;
5. enumerate the challenges of the Nigerian State towards Nation building;
6. analyse the role of the Judiciary in upholding people's fundamental rights;
7. identify acceptable norms and values of the major ethnic groups in Nigeria; and
8. list and suggest possible solutions to identifiable Nigerian environmental, moral and value problems.

MTH 102: Elementary Mathematics II (Calculus) (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. distinguish types of rules in Differentiation and Integration;

2. describe the meaning of Function of a real variable, graphs, limits and continuity; and
3. solve some applications of definite integrals in areas and volumes.

Course Contents
Function of a real variable, graphs, limits and idea of continuity. The derivative, as limit of rate of change. Techniques of differentiation. Extreme curve sketching; Integration as an inverse of differentiation. Methods of integration, Definite integrals. Application to areas, volumes.

PHY 102: General physics II (Electricity & magnetism) (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. describe the electric field and potential, and related concepts, for stationary charges;
2. calculate electrostatic properties of simple charge distributions using Coulomb's law, Gauss's law, and electric potential;
3. describe and determine the magnetic field for steady and moving charges;
4. determine the magnetic properties of simple current distributions using Biot-Savart and Ampere's law;
5. describe electromagnetic induction and related concepts and make calculations using Faraday and Lenz's laws;
6. explain the basic physical of Maxwell's equations in integral form;
7. evaluate DC circuits to determine the electrical parameters;
8. determine the characteristics of ac voltages and currents in resistors, capacitors, and Inductors.

Course Contents
Forces in nature. Electrostatics (electric charge and its properties, methods of charging). Coulomb's law and superposition. Electric field and potential. Gauss's law. Capacitance. Electric dipoles. Energy in electric fields. Conductors and insulators. DC circuits (current, voltage and resistance. Ohm's law. Resistor combinations. Analysis of DC circuits. Magnetic fields. Lorentz force. Biot-Savart and Ampère's laws. Magnetic dipoles. Dielectrics. Energy in magnetic fields. Electromotive force. Electromagnetic induction. Self and mutual inductances. Faraday and Lenz's laws. Step up and step down transformers. Maxwell's equations. Electromagnetic oscillations and waves. AC voltages and currents applied to inductors, capacitors, and resistance.

PHY 108 - General Practical Physics II (1 Unit C: PH 45)
Learning Outcomes
On completion, the student should be able to:
1. conduct measurements of some physical quantities;
2. make observations of events, collect and tabulate data;
3. identify and evaluate some common experimental errors;
4. plot and analyse graphs;
5. draw conclusions from numerical and graphical analysis of data; and
6. prepare and present practical reports.
Course Contents
This practical course is a continuation of PHY 107 and is intended to be taught during the second semester of the 100 level to cover the practical aspect of the theoretical courses that have been covered with emphasis on quantitative measurements, the treatment of measurement errors, and

graphical analysis. However, emphasis should be placed on the basic physical techniques for observation, measurements, data collection, analysis and deduction.

COS 102: Problem Solving (3 Units C: LH 30; PH 45)

Learning Outcomes

At the end of this course, students should be able to:

1. explain problem solving processes;
2. demonstrate problem solving skills;
3. describe the concept of algorithms development and properties of algorithms;
4. discuss the solution techniques of solving problem;
5. solve computer problems using algorithms, flowcharts, pseudocode; etc.; and
6. solve problems using programming language using C, PYTHON, etc.

Course Contents

Introduction to the core concepts of computing. Problems and problem-solving. The identification of problems and types of problems (routine problems and non-routine problems). Method of solving computing problems (introduction to algorithms and heuristics). Solvable and unsolvable problems. Solution techniques of solving problems (abstraction, analogy, brainstorming, trial and error, hypothesis testing, reduction, literal thinking, means end analysis, method of focal object, morphological analysis, research, root cause analysis, proof, divide and conquer). General Problem-solving process. Solution formulation and design: flowchart, pseudocode, decision table, decision tree. Implementation, evaluation and refinement. Programming in C, Python etc.

Lab Work: Use of simple tools for algorithms and flowcharts; writing pseudocode; writing assignment statements, input-output statements and condition statements; demonstrating simple programs using any programming language (Visual Basic, Python, C)

CLU-CSC 134 Web Programming 1 (3 Units: (Compulsory): 15H (Lecture) 45H (Practical)

Senate Approved Relevance

Many businesses today are conducted in online, this has made web programming abilities profitable, and so, in line with the vision and mission of Chrisland University to produce graduate with sound knowledge that can proffer solutions to societal needs, web programming is a worthy course that would make the graduates of the University to lead their peers in the society as light as they would be armed with knowledge that would enable them to produce web-based applications for diverse sector of the economy

The course seeks to expose students to the development of dynamic web site and web-based applications using HTML, CSS, JavaScript and XML.

Overview

This course is devoted to the development of Web sites with standards considering client-side programming. Students learn how the internet works, the web standards mode, web design concepts, HTML, CSS, JavaScript, XML and will be introduced to PHP.

Students will be required to carry out practical work in terms of assignments and small projects.

Objectives

The objectives of the course are to:

1. explain the design principles that relate to web design
2. describe the basic language of the web: HTML and CSS
3. give in-depth explanation of Cascading Style Sheet
4. provide an overview of some JavaScript frameworks
5. provide detailed discussion of PHP and its intricacies
6. give an overview of mobile web

Learning Outcomes

At the end of this course, students should be able to:

1. develop a functioning website
2. examine the protocols and systems used on the web.
3. identify the functions of clients and servers on the web.
4. design interactive web sites using standards for maximum usability and accessibility
5. implement basic JavaScript.

Course Contents

The Internet. Introduction to HTML. HTML tags. Cascading Styling Sheets. How to Add CSS to HTML. JavaScript. JavaScript Frameworks. How to Add JavaScript to HTML. XML structure. Document type definition. Parsed Entities. Content and Markup. Element Declaration. Attribute Declarations. Parameter Entities. Namespaces in XML. Languages and XML. Whitespace, Comments and Processing Instructions. Server Side Programming with PHP. General types in PHP. Functions. Strings. Arrays. Classes and Objects. Operators and Control Structures. Databases. Introduction to JavaScript. Introduction to Mobile Web

Minimum Academic Standards

CCMAS Benchmarks: A laboratory equipped with internet enabled capabilities and other NUC MAS requirements.

GST 102: COMMUNICATION IN ENGLISH II (2 Units)

The course is designed to address the areas of speaking, reading, comprehension and communication. There are two significant categories of topics, sound, and literary appreciation. Students would be required to read and analyse texts from the genres of literature.

GST 106: LOGIC, PHILOSOPHY AND HUMAN EXISTENCE (2 Units C: LH 30)

Learning Outcomes

At the end of this course, students should be able to:

1. know the basic features of philosophy as an academic discipline;
2. identify the main branches of philosophy & the centrality of logic in philosophical discourse;
3. know the elementary rules of reasoning;
4. distinguish between valid and invalid arguments;
5. think critically and assess arguments in texts, conversations and day-to-day discussions;
6. critically asses the rationality or otherwise of human conduct under different existential

conditions;
7. develop the capacity to extrapolate and deploy expertise in logic to other areas of knowledge, and
8. guide his or her actions, using the knowledge and expertise acquired in philosophy and logic.

Course Contents

Scope of philosophy; notions, meanings, branches and problems of philosophy. Logic as an indispensable tool of philosophy. Elements of syllogism, symbolic logic— the first nine rules of inference. Informal fallacies, laws of thought, nature of arguments. Valid and invalid arguments, logic of form and logic of content — deduction, induction and inferences. Creative and critical thinking. Impact of philosophy on human existence. Philosophy and politics, philosophy and human conduct, philosophy and religion, philosophy and human values, philosophy and character molding, etc.

GST 110: CONTEMPORARY HEALTH ISSUES (2 Units)

The purpose of this course is to introduce students to concepts of health and wellness. Diet, exercise, diseases, drugs and substance use are all issues affecting human lives. This course presents an overview of these issues and provides basic knowledge on wellness and health as concepts. Students will also familiarize themselves with current illnesses and associated issues.

CLU-SEG 203 BLOCKCHAIN TECHNOLOGY (3 Units: (Compulsory): 15 H (Lecture), 45 H (Practical)

Senate Approved Relevance

Blockchain improves the traceability, security, trustworthiness, and transparency of data shared across a business network while generating new efficiencies that save costs. Blockchain for business employs an open, unchangeable ledger that only members with authorization can view. When trust has become a major challenge in business and every organization in Nigeria, it is important for Chrisland university students who are being trained to show forth the light to develop a technique to solve such problem in our society.

Overview

It's common to link crypto currencies like Bitcoin and others to blockchain technology. However, it is a cutting-edge technology that includes developments in information systems design and cryptography. Information security and privacy have come to the forefront of discussion in today's environment of ever-increasing volumes of data. The business applications of blockchain technology are the main topic of this program. It encompasses the full knowledge spectrum, ranging from the technology's basic characteristics to cutting-edge business concepts built on top of it. Students will gain knowledge about how to recognize the blockchain's potential and constraints in various business settings during this course. We'll go over many techniques that companies can utilize to upgrade their data management.

Objectives

The objectives of the course are to:

1. discuss the depth of blockchain technology.
2. explain the implementation of blockchain.
3. describe the use of blockchain in cryptocurrency.
4. describe the areas of application of blockchain.
5. expose the students to various implementations of blockchain technology such as bitcoin, Ethereum, and Hyper ledger

Learning outcomes

At the end of this course, student should be able to:

1. Understand all the basis of blockchain and the cryptographic technology.
2. Categorize the different types of blockchain and where it is appropriate for use.
3. Understand and identify blockchain use cases.
4. Develop the ability to formulate mathematical models and problem-solving skills through programming techniques for addressing real-time problems using appropriate data structures and algorithms.
5. Develop the ability to provide design, build, and deploy a distributed application and provide solutions using blockchain applications to enhance business measures by sharing information securely.
6. Understand how to create cryptocurrencies and give a strong technical understanding of blockchain technologies with an in-depth understanding of applications, research challenges, and future directions.
7. Apply knowledge of implementations of Bitcoin, Ethereum and Hyperledger to develop solutions in the appropriate domains.

Course contents

Introduction to blockchain technology. Distributed Systems. The History of Blockchain. CAP theorem and blockchain. Benefits and Limitations of Blockchain. Decentralisation using Blockchain. Methods of Decentralisation. Routes to Decentralisation. Cryptography in Blockchain. Cryptographic primitives. Asymmetric cryptography. Public and Private Keys. Categories of blockchain. Blockchain Architecture Design and Use Cases. Distributed ledger. Blockchain Ecosystem. Areas of application of blockchain**.** Cryptocurrency technologies. Introduction to BitCoin. Transactions and Structure of BitCoin. The BitCoin Network. Ethereum. Hyperledger.

Minimum Academic Standards

CCMAS Benchmarks: A standard computer laboratory equipped with internet enabled capabilities, an android mobile phone and a simulation laboratory may be necessary. Other NUC MAS requirements.

CLU-SEG 231 Introduction to Green Computing (3 Units: (Elective): 45H (Lecture) 15H (Practical)

Senate Approved Relevance

Training of graduates who are highly skilled in not only conceptualization, development and deployment of IT based solutions to solve human life in their academic endeavour but also who can promote the approach to manufacture eco-friendly hardware devices and components, innovative techniques for developing energy saving software(s) and set of practices to support the critical mission “Saving Earth”. The course will equips the students the awareness on steps to follow to play important role in developing computing in green environment and therefore shine the light forth as contained in the vision and mission statement of the university as embedded in the motto of the school “shine forth the Light”. Relevance is seen in CLU desire to have a safe environment in all facets of human life that would be a beacon that shines the light for people to follow considering the current economic realities and cost of energy.

Overview

The planet “Earth” is facing problem of global warming and greenhouse gas (GHG) emission because of industrial development all over the world where IT plays the master role to integrate modern technologies and systems. The researches show that energy consumption by IT resources is increasing day by day and this is resulting in environmental pollution and economic loss. To avert this, new paradigm called "Green Computing" coined, i.e. an approach to manufacture eco-friendly hardware devices & components, innovative techniques for developing energy saving software(s) and set of practices to support the critical mission “Saving Earth”.

The course is designed to expose the student to relevant knowledge that will impacts environmental friendliness, saving powers, long term profit, reduce pollution, power management and increase performance. It also disseminates awareness on steps to follow to play important role in developing computing in green environment.

Objectives:

The objectives of the course are to:

1. explain historical development of green computing;
2. define green computing;
3. state chemical elements used in computer;
4. explain environmental concerns and sustainable development;
5. explain different environment regulations;
6. explain some function green computing;
7. explain the green computing dimension;
8. explain the green computing life cycle;
9. explain the techniques for green computing;
10. explain the characteristics of green computing;
11. explain the green washing;

12. state applications of green computing;
13. explain green computing awareness;
14. explain the challenges of green computing.

Learning outcomes

On completion of the course, students should be able to:

1. explain historical development of green computing;
2. define green computing;
3. apply at least one strategy of green computing to their software development;
4. state 5 characteristics of green computing;
5. compare at least three environmental regulations and come up with the best for chrisland university community;
6. state 3 dimensions green computing;
7. differentiate identify the infusion of green washing in green computing awareness;
8. disseminate the green computing awareness to other students;
9. states at least five impacts of green computing;
10. Identify at least 6 challenges facing green computing.

Course Contents

Fundamentals of green computing. Historical development of green computing. Green computing paradigm. Chemical element used in computer. Environmental concerns and sustainable development. Environment regulations. Why green computing. Green computing Dimension. Green computing framework. Greening computer's entire life cycle. The three R's of green computing. Techniques for green computing. Characteristics of green computing. Enterprise green it strategy. Green washing. Green it: burden or opportunity? Application of green computing. Impacts of green computing. Green computing awareness. Issues & challenges of green computing

Minimum Academic Standards

CCMAS Benchmarks: A laboratory/studio equipped with computing resources.

GST 204: PEACE AND CONFLICT RESOLUTION (2 Units C: LH 30)

Learning Outcomes

At the end of the course, students should be able to:
1. analyse the concepts of peace, conflict and security;
2. list major forms, types and root causes of conflict and violence;
3. differentiate between conflict and terrorism;
4. enumerate security and peacebuilding strategies; and
5. describe roles of international organisations, media and traditional institutions in peace building.

Course Contents

Concepts of Peace, Conflict and Security in a multi-ethnic nation. Types and Theories of Conflicts: Ethnic, Religious, Economic, Geopolitical Conflicts; Structural Conflict Theory, Realist Theory of Conflict, Frustration-Aggression Conflict Theory. Root causes of Conflict and Violence in Africa: Indigene and Settlers Phenomenon; Boundaries/border disputes; Political disputes; Ethnic disputes and rivalries; Economic Inequalities; Social disputes; Nationalist Movements and Agitations; Selected Conflict Case Studies – Tiv-Junkun; Zango Kartaf, Chieftaincy and Land disputes, etc. Peace Building, Management of Conflicts and Security: Peace & Human Development. Approaches to Peace & Conflict Management (Religious, Government, Community Leaders, etc.). Elements of Peace Studies and Conflict Resolution: Conflict dynamics assessment Scales: Constructive & Destructive. Justice and Legal framework: Concepts of Social Justice; The Nigeria Legal System. Insurgency and Terrorism. Peace Mediation and Peace Keeping. Peace & Security Council (International, National and Local levels) Agents of Conflict resolution – Conventions, Treaties Community Policing: Evolution and Imperatives. Alternative Dispute Resolution, ADR. Dialogue b). Arbitration, c). Negotiation d). Collaboration, etc. Roles of International Organisations in Conflict Resolution. (a). The United Nations, UN and its Conflict Resolution Organs. (b). The African Union & Peace Security Council (c). ECOWAS in Peace Keeping. Media and Traditional Institutions in Peace Building. Managing Post-Conflict Situations/Crisis: Refugees. Internally Displaced Persons, IDPs. The role of NGOs in Post-Conflict Situations/Crisis.

GST 205: ENTREPRENEURSHIP STUDIES I (2 Units C: LH 15; PH 45)

Entrepreneurship Studies I exposes students to the fundamental principles of entrepreneurship and Nigeria's entrepreneurship ecosystem.

Learning Outcomes

At the end of this course, students should be able to:

1. explain the concepts and theories of entrepreneurship, intrapreneurship, opportunity seeking, new value creation, and risk taking;
2. state the characteristics of an entrepreneur;
3. analyse the importance of micro and small businesses in wealth creation, employment, and financial independence;
4. engage in entrepreneurial thinking;
5. identify key elements in innovation;
6. describe stages in enterprise formation, partnership and networking including business planning;
7. describe contemporary entrepreneurial issues in Nigeria, Africa and the rest of the world; and
8. state the basic principles of e-commerce.

Course Contents

Concept of Entrepreneurship (Entrepreneurship, Intrapreneurship/Corporate Entrepreneurship). Theories, Rationale and relevance of Entrepreneurship (Schumpeterian and other perspectives, risk-taking, necessity and opportunity-based entrepreneurship and creative destruction). Characteristics of Entrepreneurs (Opportunity seeker, risk taker, natural and nurtured, problem solver and change agent, innovator and creative thinker). Entrepreneurial thinking (Critical thinking, Reflective thinking, and Creative thinking). Innovation (Concept of innovation,

Dimensions of innovation, Change and innovation, Knowledge and innovation). Enterprise formation, partnership and networking (Basics of business plan, Forms of business ownership, business registration and forming alliances and joint ventures). Contemporary Entrepreneurship Issues (knowledge, skills and technology, intellectual property, virtual office, networking). Entrepreneurship in Nigeria (Biography of inspirational entrepreneurs, youth and women entrepreneurship, Entrepreneurship support institutions, Youth enterprise networks and environmental and cultural barriers to entrepreneurship). Basic principles of e-commerce.

GST 207: NIGERIA'S PEOPLE, CULTURE, GOVERNMENT AND ECONOMY (2 Units C: LH 30)

The course shall introduce students to the background and evolution of Nigerian State. The course shall also give students the basic understanding of the socio-cultural values of Nigeria, in terms of governance, economic, and socio-cultural values.

The course will include lectures cum research and term paper discussion, which are assigned to students: Oral and written presentation of assignments, library research, teamwork and small group presentations, and other activities. Students are required to come to class, to participate in scheduled activities. They will also be encouraged to share experiences, knowledge, talents, and skills, which are related to the entire philosophy of the course.

Learning Outcomes

At the end of the course, students should be able to:

1. analyse the historical foundation of the Nigerian culture and arts in pre-colonial times;
2. list and identify the major linguistic groups in Nigeria;
3. explain the gradual evolution of Nigeria as a political unit;
4. analyse the concepts of Trade, Economic and Self-reliance status of the Nigerian peoples towards national development;
5. enumerate the challenges of the Nigerian State towards Nation building;
6. analyse the role of the Judiciary in upholding people's fundamental rights;
7. identify acceptable norms and values of the major ethnic groups in Nigeria; and
8. list and suggest possible solutions to identifiable Nigerian environmental, moral and value problems.

GST 211: INSTRUMENTATION IN MUSIC (0 Unit)

Students will learn how to play an instrument of their choice including voice.

MTH 201: Mathematical Methods I (2 Units C: LH 30)

Learning Outcomes

At the end of the course students should be able to:

1. describe Real-valued functions of a real variable;
2. solve some problems using Mean value Theorem and Taylor Series expansion; and
3. evaluate Line Integral, Surface Integral and Volume Integrals.

Course Contents

Real-valued functions of a real variable. Review of differentiation and integration and their applications. Mean value theorem. Taylor series. Real-valued functions of two and three

variables. Partial derivatives chain rule, extrema, Lagrangian multipliers. Increments, differentials and linear approximations. Evaluation of line, integrals. Multiple integrals.

COS 201: Computer Programming I (3 Units C1: LH 30; PH 45)

Learning Outcomes

At the end of this course, students should be able to:

1. identify different programming paradigms and their approaches to programming;
2. write programmes using basic data types and strings;
3. design and implement programming problems using selection;
4. design and implement programming problems using loops;
5. use and implement classes as data abstractions in an object-oriented approach;
6. implement simple exception handling in programmes;
7. develop programmes with input/output from text files; and
8. design and implement programming problems involving arrays.

Course Contents

Introduction to computer programming. Functional programming; Declarative programming; Logic programming; Scripting languages. Introduction to object-orientation as a technique for modelling computation. Introduction of a typical object-oriented language, such as Java. Basic data types, variables, expressions, assignment statements and operators. Basic objectoriented concepts: abstraction; objects; classes; methods; parameter passing; encapsulation. Introduction to Strings and string processing; Simple I/O; control structures; Arrays; Simple recursive algorithms; inheritance; polymorphism.

Lab work: Programming assignments involving hands-on practice in the design and implementation of simple algorithms such as finding the average, standard deviation, searching and sorting. Practice in developing and tracing simple recursive algorithms. Developing programmes involving inheritance and polymorphism.

SEN 201: Introduction to Software Engineering (2 units C: LH 30)

Learning Outcomes

At the end of this course, students should be able to:

1. describe the concept of the software life cycle;
2. explain the phases of requirements analysis, design, development, testing and maintenance in a typical software life cycle;
3. differentiate amongst the various software development models;
4. utilise UML for object-oriented analysis and design;
5. describe different design architectures;
6. explain the various tasks involved in software project management; and
7. describe the basic legal issues related to Software Engineering.

Course Contents

Software Engineering concepts and principles. Design, development and testing of software systems. Software processes: software lifecycle and process models. Process assessment models. Software process metrics. Life cycle of software system. Software requirements and

specifications. Software design. Software architecture. Software metrics. Software quality and testing. Software architecture. Software validation. Software evolution: software maintenance; characteristics of maintainable software; re-engineering; legacy systems; software reuse. Software Engineering and its place as a computing discipline. Software project management: team management; project scheduling; software measurement and estimation techniques; risk analysis; software quality assurance; software configuration management. Software Engineering and law.

CSC 203: Discrete Structures (2 Units C: LH 30)

Learning Outcomes

At the end of this course, the students will be able to:

1. convert logical statements from informal language to propositional and predicate logic expressions;
2. describe the strengths and limitations of propositional and predicate logic;
3. outline the basic structure of each proof technique (direct proof, proof by contradiction, and induction) described in this unit;
4. apply each of the proof techniques (direct proof, proof by contradiction, and induction) correctly in the construction of a sound argument;
5. apply the pigeonhole principle in the context of a formal proof;
6. compute permutations and combinations of a set, and interpret the meaning in the context of the particular application;
7. map real-world applications to appropriate counting formalisms, such as determining the number of ways to arrange people around a table, subject to constraints on the seating arrangement, or the number of ways to determine certain hands in cards (e.g., a full house); and
8. solve a variety of basic recurrence relations.

IFT 211: Digital Logic Design (2 Units C: LH 15; PH 45)

Learning Outcomes

At the end of this course, students will be able to:

1. explain why everything is data, including instructions, in computers;
2. describe how negative integers, fixed-length numbers, and non-numeric data are represented;
3. convert numerical data from one format to another;
4. describe computations as a system characterised by a known set of configurations with transitions from one unique configuration (state) to another (state);
5. describe the distinction between systems whose output is only a function of their input (combinational) and those with memory/history (sequential);
6. describe a computer as a state machine that interprets machine instructions;
7. articulate that there are many equivalent representations of computer functionality, including logical expressions and gates, and be able to use mathematical expressions to describe the functions of simple combinational and sequential circuits; and
8. design the basic building blocks of a computer: arithmetic-logic unit (gate-level), registers (gate-level), central processing unit (register transfer-level), and memory (register transfer-level).

Course Contents

Introduction to information representation and number systems. Boolean algebra and switching theory. Manipulation and minimisation of completely and incompletely specified Boolean functions. Physical properties of gates: fan-in, fan-out, propagation delay, timing diagrams and tri-state drivers. Combinational circuits design using multiplexers, decoders, comparators and adders. Sequential circuit analysis and design, basic flip-flops, clocking and timing diagrams. Registers, counters, RAMs, ROMs, PLAs, PLDs, and FPGAs.

Lab Work: Simple combinational gates (AND, OR, NOT, NAND, NOR); Combinational circuits design using multiplexers, decoders, comparators and adders. Sequential circuit analysis and design using basic flip-flops (S-R, J-K, D, T flip-flops); Demonstration of registers, counters, RAMs, ROMs, PLAs, PLDs, and FPGAs.

GST 205: ENTREPRENEURSHIP STUDIES I (2 Units)

Entrepreneurship Studies I exposes students to the fundamental principles of entrepreneurship and Nigeria's entrepreneurship ecosystem.

GST 207: NIGERIA'S PEOPLE, CULTURE, GOVERNMENT AND ECONOMY(2 Units)

The course shall introduce students to the background and evolution of Nigerian State. The course shall also give students the basic understanding of the socio-cultural values of Nigeria, in terms of governance, economic, and socio-cultural values.

The course will include lectures cum research and term paper discussion, which are assigned to students: Oral and written presentation of assignments, library research, teamwork and small group presentations, and other activities. Students are required to come to class, to participate in scheduled activities. They will also be encouraged to share experiences, knowledge, talents, and skills, which are related to the entire philosophy of the course.

GST 211: INSTRUMENTATION IN MUSIC (0 Unit)

Students will learn how to play an instrument of their choice including voice.

CLU-SEG 208 System and Software Specification and Requirements (3 Units: (Compulsory): 15H (Lecture) 45H (Practical)

Senate Approved Relevance

System and software specification and requirements are essential for a successful software and system development. Without a clear understanding of the requirements, it is challenging and infeasible to develop software or systems that meet the needs of end-users and stakeholders, delivered on time and within budget, and are of high quality.

In line with the universities core values of research and intellectual freedom, this course is critical to the success of software and system development projects. The specification and

requirements process involves identifying, analysing, documenting, validating, and managing the needs of users and stakeholders.

Overview

The requirements process is a critical component of software and system development, as it helps to ensure that software and systems are developed to meet the needs of users and stakeholders. This process also helps to reduce project risks, improve communication and collaboration, and enhance the quality of the final product.

System and software specification and requirements refer to the process of identifying, analysing, documenting, validating, and managing the needs of end-users and stakeholders in software and system development projects. It is essential for developing software and systems that meet the needs of users, are of high quality, and are delivered on time and within budget.

This course is designed to provide students with the skills and knowledge needed to identify, analyse, document, validate, and manage requirements for software and system development projects. It may also include practical exercises and projects to help students apply the concepts and techniques learned in class.

By the end of the course, students should have a solid understanding of the system and software specification and requirements process and be able to apply these skills in real-world software and system development projects.

Objectives

The objectives of the course are to:

12. develop an understanding of the importance of requirements engineering in software and system development.
13. acquire the skills and techniques needed to identify, analyse, document, validate, and manage requirements effectively.
14. understand the different types of requirements and their role in the development process.
15. learn how to elicit requirements from stakeholders, including end-users, customers, and domain experts.
16. understand how to validate and verify requirements to ensure that they meet the needs of end-users and stakeholders.
17. improve communication and collaboration with stakeholders, including end-users, developers, and project managers.
18. learn how to document requirements using different tools and techniques, such as use cases, user stories, and requirements traceability matrices.
19. develop the ability to apply the requirements engineering process in real-world software and system development projects.

Learning outcomes

1. Demonstrate an understanding of the importance of requirements engineering in software and system development.
2. Use different elicitation techniques to identify and analyse requirements from various stakeholders.
3. Evaluate requirements for completeness, consistency, and feasibility.
4. Develop a validation plan and execute it to ensure that requirements are met.
5. Identify and mitigate risks associated with requirements development and management.
6. Analyse and compare different requirements engineering methods and tools to select the appropriate ones for specific software and system development projects.
7. Apply the requirements engineering process in real-world software and system development projects.
8. Describe and document requirements using different tools and techniques, such as use cases, user stories, and requirements traceability matrices.

Course Contents

Introduction to System and Software Requirements. Definition, Types, Importance of Requirements in Software and System Development. The Requirements Engineering Process. Software requirements Elicitation: Techniques, Methods, Evaluation Criteria.

Requirements Analysis and Specification. Requirement Analysis Process. Functional and Non-Functional Requirements. Use Case Modelling. Entity Relationship Modelling. Data Flow Diagramming. Requirements Specification Document. Requirements Validation and Verification. Requirements Review. Requirements Testing. Traceability Matrix. Requirements Management. Requirements Traceability. Change Management. Configuration Management. Requirements Prioritization. Case Studies and Best Practices.

Minimum Academic Standards

CCMAS Benchmarks: A laboratory equipped with internet enabled capabilities. Other NUC MAS requirements.

CLU-SEG 222 Software Engineering (3 Units): (Compulsory): 30H (Lecture), 30H (Practical)

Overview

Engineering is an ever-evolving field, including software engineering. It deals with methodical strategies used by programming teams to construct sizable software systems. The key components of software engineering, including product-related concerns like requirements, design, validation, and process-related ones like process models and their evaluation, have been briefly reviewed. Therefore, there is need for the study of such a course in the Nigerian University system.

Senate Approved Relevance

The significance of software engineering is only going to increase given how widespread software is in modern life. Software controls a growing number of technologies in a wide range of fields, which presents challenges, demands, and obligations for software developers. For instance, the expansion of the Internet has increased the demand for novel approaches to the creation and extensive use of software systems and products. The creation of methods for achieving security in software systems was required by the development of e-commerce applications. Software engineering promises to remain a dynamic and active field in a perpetual state of change as new applications and technology are always developing. It thereby becomes necessary in our curriculum which will also create opportunities for our student in the industry, as Chrisland University prides herself as promoting the entrepreneurship skills in her students.

Objectives

The objectives of the course are to:

1. introduce the essential software engineering concepts that are commonly used.
2. develop skills useful for software development.
3. explain software engineering practices and standards.
4. identify the basic steps in software engineering and development.
5. explain dependability and security features of software

Expected learning outcome

At the end of this course, student must be able:

1. To apply the principles of software engineering process in software development.
2. To understand planning, scheduling and estimation in a software development project.
3. To model the requirement that is needed in a software project.
4. To design and test the requirement in a software project.
5. To identify and implement software development cycle.
6. Describe the security and dependability essentials of software

Course content

Overview of software engineering. Software Engineering Ethics. Introduction to software project management. Software development cycle. Software Process Activities. The Rational Unified Process. Agile Software Development. Modelling requirement. Software design. Software implementation. Software Testing. Software validation and verification. Software evolution. Quality assurance. Dependability and security of software. Dependability and security specification. Security engineering. Dependability and security assurance. Recent trends in software development.

Minimum Academic Standards

CCMAS Benchmarks: A standard software laboratory. Other NUC MAS requirements.

MTH 202: Mathematical Methods II (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. define the following: order and degree of a differential equation;
2. describe some techniques for solving first and second order linear and non-linear equations; and
3. solve some problems related to geometry and physics.

Course Contents
Derivation of differential equations from primitive, geometry, physics etc. order and degree of differential equation. Techniques for solving first and second order linear and non-linear equations. Solutions of systems of first order linear equations. Finite linear difference equations. Application to geometry and physics.

COS 202: Computer Programming II (3 Units C: LH 30; PH 45)
Learning Outcomes
At the end of this course, students should be able to:
1. demonstrate the principles of good programming and structured programming concepts;
2. demonstrate string processing, internal searching, sorting, and recursion;
3. demonstrate the basic use of OOP concepts: classes, objects, inheritance, polymorphism, data abstraction;
4. apply the tools for developing, compiling, interpreting and debugging programs; and
5. demonstrate the use of syntax and data objects, operators. Central flow constructs, objects and classes programming, Arrays, methods, Exceptions, Applets and the Abstract, OLE, Persistence, Window Toolkit.

Course Contents
Review and coverage of advanced object-oriented programming - polymorphism, abstract classes and interfaces; Class hierarchies and program organisation using packages/namespaces; Use of API – use of iterators/enumerators, List, Stack, Queue from API; Searching; sorting; Recursive algorithms; Event-driven programming: event-handling methods; event propagation; exception handling. Applications in Graphical User Interface (GUI) programming.

Lab work: Programming assignments leading to extensive practice in problem solving and program development with emphasis on object-orientation. Solving basic problems using static and dynamic data structures. Solving various searching and sorting algorithms using iterative and recursive approaches. GUI programming.

INS 204: Systems Analysis and Design (3 Units C: LH 30; PH 45)
Learning Outcomes
At the end of this course, students should be able to:
1. describe system requirements gathering techniques;
2. explain data modelling technique (entity relationship modelling);
3. explain process modelling technique (data flow diagram);
4. describe system architectural design;

5. describe process and database design; and
6. explain user interface design.

Course Contents

Structured approach to analysis and design of information systems for businesses. Software development life cycle. Structured top-down and bottom-up design. Dataflow diagramming. Entity relationship modelling. Computer aided software engineering. Input and output, prototyping design and validation. File and database design. Design of user interfaces. Comparison of structured and object-oriented design

Lab Work: system requirements gathering techniques;data modelling techniques (entity relationship modelling);process modelling techniques (data flow diagram); use of UML diagrams; system architectural design; user interface design.

IFT 212: Computer Architecture and Organisation (3 Units C; LH 30; PH 45)

Learning Outcomes:

At the end of this course, student should be able to:
1. explain different instruction formats, such as addresses per instruction and variable length vs. fixed length formats;
2. describe the organisation of the classical von Neumann machine and its major functional units;
3. explain how subroutine calls are handled at the assembly level;
4. describe the basic concepts of interrupts and I/O operations;
5. write simple assembly language program segments;
6. show how fundamental high-level programming constructs are implemented at the machine-language level;
7. compare alternative implementation of data paths;
8. discuss the concept of control points and the generation of control signals using hardwired or micro-programmed implementations;

Course Contents

Instruction format and types, memory and I/O instructions, dataflow, arithmetic, and flow control instructions, addressing modes, stack operations, and interrupts. Data path and control unit design. RTL, microprogramming, and hardwired control. Practice of assembly language programming. Memory hierarchy, cache memory, virtual memory. I/O fundamentals. Interrupt structures.

Lab work: Programming assignments to practice MS-DOS batch programming, Assembly Process, Debugging, Procedures, Keyboard input, Video Output, File and Disk I/O and Data Structure. Instruction and arithmetic pipelining, superscalar architecture. Reduced Instruction Set Computers. Parallel architectures and interconnection networks.

GST 206: ENTREPRENEURSHIP STUDIES II (2 Units)

Profiles of business ventures in the various business ventures such as: Soap/Detergent, Tooth brush and Tooth paste making; Photography; Brick making; Rope making; Brewing; Glassware production/Ceramic production, Paper production; Water treatment/conditioning/packaging; Food processing/preservation/packaging; Metal fabrication; Tanning industry; Vegetable oil extraction; Farming; Fisheries/Aquaculture; Plastic making; Refrigeration/air-conditioning; Carving; Weaving; Bakery; Tailoring; Printing; Carpentry, Interior Decoration; Animal husbandry, etc. Case study methodology applied to the development and administration of cases that bring out key issues of business

environment, start-up, pains, and gains of growth of businesses, etc, with particular reference to Nigerian businesses. Experience sharing by business actors in the economy with students during case presentations.

GST 208: COMMUNICATION IN FRENCH I (1 Unit)

The communication objectives of the course include getting acquainted with somebody: Introducing oneself, Apologizing, Saying one's name, Asking someone for her/his name. The tasks to be realised in the course amongst others pertain to meeting and greeting people, creating a contact list for the class, discovering French alphabet, identifying some objects using politeness formula.

SEG 311 Software Project Management I (3 Units: (Compulsory): 15H (Lecture) 45H (Practical)

Senate Approved Relevance

Educating students on Software Project Management equips them with the much sought after knowledge of relevance in the field of Software Engineering. It gives students the needed edge which comes with being informed about trending matters. Software Project Management involves planning, organizing, monitoring, and controlling the software development process to ensure the completion of a high-quality software product within the specified time, budget, and scope. The goal of software project management is to deliver a software product that meets the customer's needs and expectations, and to do so efficiently and effectively.

Applying the knowledge gained on software project management on any societal problem will make the solution from such endeavour enduring, thereby making the solution provider "Shine forth the Light" as contained in the mission and vision of the university.

With the recent push for online courses, digital lesson plans, and remote learning, Software Project Management is more important now than ever. The knowledge of the Software Project Management enables student's transverse the realm of Project Management with ease.

Overview

Software Project Management is the process of planning, executing, and closing the work of a team to achieve specific goals and meet success criteria at a specified time. The primary challenge of software project management is to achieve all of the project goals within the given constraints, which typically include limited time, budget, and resources.

The goal of software project management is to create a software product that meets the needs of the users and adds value to the organization, while also ensuring that the project is completed efficiently and effectively, with all stakeholders working together towards a common goal.

In addition to delivering a high-quality software product, successful software project management also requires good communication, collaboration, and teamwork among the project stakeholders. By fostering a positive, productive working environment, a software project manager can help ensure that the project is completed smoothly and that all parties involved are satisfied with the outcome.

Objectives

The objectives of the course are to:

1. introduce the students to the concept of Software Project Management.
2. deliver a high-quality software product that meets the needs and expectations of the stakeholders.
3. help students understand the purpose of Software Project Management.
4. enable the student, communicate effectively with stakeholders, including clients, and project team members to ensure that everyone is aware of project progress and any potential issues that may arise.
5. teach students how to effectively manage and allocate resources, including personnel, equipment, and materials, to ensure the project is completed efficiently and effectively.
6. expose the students to the basis of keeping the balance between the theoretical foundations and practical consideration.
7. establish and maintain high-quality standards, and ensure that the final software product meets these standards.
8. ensure that the project is progressing according to plan and to make necessary adjustments as needed.

Learning outcomes

1. To make the student know various project management frameworks such as Agile, Waterfall, Scrum, etc., and their strengths and limitations.

2. To enlighten the student on the ability to continuously evaluate and improve processes, systems, and practices to ensure the success of future projects.

3. To be able to effectively manage changes that occur during a project's lifecycle.

4. Students will be able to analyze and evaluate software systems to identify areas for improvement and make recommendations for future development.

5. Students will learn the importance of risk management which includes the ability to identify, assess, and manage potential risks that could impact a project's success.

6. The students will be able to understand the project management methodologies.

7. To make the student have the ability to create, manage and monitor a project budget, and ensure that project costs are kept within budget constraints.

8. To establish and maintain quality standards and ensure that the project meets the desired quality levels.

Course Contents

Software Project Management Complexities. Responsibilities of Software Project Manager. Project Estimation Techniques. Project Planning and Scheduling. Organization and Team Structures. Risk Management. Managing People. Team work. Software Configuration Management. Quality management. Software measurement and metrics. Measurement ambiguities. Software component analysis. Process improvement. Process measurement. Process analysis. Process change.

Minimum Academic Standards

CCMAS Benchmarks: A laboratory equipped with internet enabled capabilities. Other NUC MAS requirements.

CLU-SEG 323 Foundation to Machine Learning (3 Units: Compulsory): 30H (Lecture) 30H (Practical)

Senate Approved Relevance

The Foundation of Machine Learning provides comprehensive coverage of both learning algorithms: supervised and unsupervised learning. Outlines the computation paradigm for solving classification, regression, and clustering problems, and it features essential techniques for building the new generation of machine learning applications; which provides solution to a wide range problem across divergent domain; a course of this magnitude would empower Chrisland University students with relevant knowledge to drive innovation in the ever dynamic technology-driven world, thereby making them financially independent as they proffer solutions that are well sought after because machine learning is now finding an application in almost every field. As a result, it makes for a great career option.

Overview

This course introduces the fundamental concepts and methods of machine learning, including the description and analysis of several modern algorithms, their theoretical basis, and the illustration of their applications. Many of the algorithms described have been successfully used in text and speech processing, bioinformatics, and other areas in real-world products and services.

Objectives

The objectives of the course are to:

1. explain the fundamentals of Machine Learning
2. explain data pre-processing

3. identify the categories of machine learning algorithms
4. describe machine learning algorithms for classification
5. describe machine learning algorithms for regression
6. describe machine learning algorithms for clustering
7. describe machine learning algorithms for association rule learning
8. examine the basic concepts of dimensionality reduction
9. explain performance metrics
10. explain model evaluation and improvement

Learning Outcomes

Students should be able to adapt to both the theoretical and practical aspects of the course such as:

1. understanding the various learning algorithms
2. understanding the basic foundation of Machine Learning
3. understanding the application areas of different machine learning algorithms
4. having insight into the concept of dimensionality reduction
5. understanding performance metrics
6. understanding model evaluation and how to improve models.

Course Contents:

The Machine Learning Landscape. What is Machine Learning. Applications of Machine Learning. Major Issues in Machine Learning. Overview of Data Preprocessing. Machine Learning Algorithms. Learning Algorithms for Classification. Basic Concepts of Classification. Types of Learners in Classification. Machine Learning Algorithms for Regression. Basic Concepts of Regression Analysis. Machine Learning Algorithms for Clustering. Cluster Analysis: Basic Concept and Methods. Requirement for Cluster Analysis. Machine Learning Algorithms for Association Rule Learning. Basic Concept of Association Rule Learning. Basic Concepts of Dimensionality Reduction. Performance Metrics. Performance Metrics for Classification Problem. Performance Metrics for Regression Problems. Model Evaluation and Improvement. Cross Validation. Grid Search

Minimum Academic Standards

CCMAS Benchmarks: A laboratory equipped with internet enabled capabilities. Other NUC MAS requirements.

SEG 331 Software Defect Management (3 Units: (Compulsory): 15H (Lecture) 45H (Practical)

Senate Approved Relevance

Defect-free software is a hallmark of quality as the prevention, finding and fixing of defects is part of quality assurance. Managing defects in software is crucial and it involves the analysis, categorization, fixing defects as well as mechanisms for preventing defects. It is therefore highly essential that students in the software/IT field are educated earlier on to understand why and how bugs/defects occur in software and also to illustrate how defect analysis techniques and approaches help resolve quality issues.

In line with the university's core values of research and intellectual freedom, this course offers a solid foundation for exploring Software Defect Management which is continually gaining prominence in the software industry. The course teaches, in detail, the analysis, importance, techniques, classification, and activities of defect management in compliance with defined standards for customer satisfaction.

Overview

To put it simply, Software Defect Management is the process of identifying, tracking, and resolving defects or bugs in software products. It's a crucial part of software development and maintenance, as the goal is to ensure that software operates as intended and meets customer requirements. Effective software defect management requires collaboration between different teams, including development, testing, and support. It also requires the use of tools and processes to ensure that defects are tracked and resolved efficiently.

By understanding the root-causes of defects found in similar projects, software processes can be improved. Improved processes, in turn, should prevent those defects from reoccurring. As a consequence, the quality of future systems can be improved efficiently and effectively.

This course is designed to cover the fundamental concepts of software defect management, from defect prevention, through to nature of software defects, root-cause analysis, techniques for management, and categorization of defects.

Objectives

The objectives of the course are to:

1. lay a solid foundation for understanding the concepts of Software Defect Management.
2. help students understand the nature of software defects and why they occur.
3. give an understanding on defect leakage and why its prevention is important
4. educate students on the techniques for root-cause analysis.
5. introduce the students to techniques for defect categorization.
6. help students understand approaches to software defect analysis.
7. give supporting knowledge on testing, defects and software quality.

8. evaluate issues in defect prevention

Learning Outcomes

1. Understanding of the basic concepts of Software Defect Management.
2. To identify the natures of software defects and why they occur.
3. Understanding on defect leakage and why its prevention is important.
4. Ability to identify techniques for analysing the root-causes of defects.
5. Ability to identify techniques for defect categorization.
6. Understanding different approaches to analysing software defects.
7. To gain more knowledge on testing, defects and software quality.
8. Ability to evaluate issues that arise in defect prevention.

Course Contents

Introduction to software defect management. Software defect management cycle. Defect states available in defect lifecycle. Defect management process. Software defect detection and identification. Classifying and Analysing software defect. Defect prioritisation and triage. Activities performed during defect triage process. Steps in defect triage process. Roles and responsibilities of participants of defect triage process. Defect triage report format. Defect resolution process. Process improvement in defect management process. Stages of defect management process. Defects and Software Quality. Defect tracking and reporting. Defect Prevention and Decision-making activities in Software Projects.

Minimum Academic Standards

CCMAS Benchmarks: A laboratory equipped with internet enabled capabilities. Other NUC MAS requirements.

SEN 301: Object-Oriented Analysis and Design (2 Units C: LH 15; PH 45)
Learning Outcomes
At the end of this course, students should be able to:
1. explain the concept of the object-oriented approach to modelling;
2. describe the conceptual model of the UML-based software development life cycle;
3. demonstrate how to use the major UML diagrams for object-oriented analysis and design;
4. demonstrate the use of UML-based CASE tools.
Course Contents
Object-oriented approach to information system development, particularly in reference to the earlier stages of analysis and design. Importance of modelling, principles of modelling, object-oriented modelling, conceptual model of the Unified Modelling Language (UML), architecture, software development life cycle. The principles and basic concepts of object orientation and the

different aspects of object-oriented modelling as represented by the UML technique. Case study of a typical UML-based CASE tool.

Lab Work: Practical exercises on different requirements specification and design activities; developing problem statements, SRS documents and Use Case Diagrams; designing UML Activity diagrams, UML Class diagrams and State Chart diagrams; drawing partial layered, logical architecture diagram with UML package diagram notation; Designing Component and Deployment diagrams.

CSC 301: Data Structures (3 Units C: LH 30, PH 45)

Learning Outcomes

At the end of this Course, students should be able to:

1. discuss the appropriate use of built-in data structures;
2. apply object-oriented concepts (inheritance, polymorphism, design patterns, etc.) in software design;
3. implement various data structures and their algorithms, and apply them in implementing simple applications;
4. choose the appropriate data structure for modelling a given problem;
5. analyse simple algorithms and determine their efficiency using big-O notation; and
6. apply the knowledge of data structures to other application domains like data compression and memory management.

Course Contents

Primitive types, Arrays, Records Strings and String processing, Data representation in memory, Stack and Heap allocation, Queues, TREES. Implementation Strategies for stack, queues, trees. Run time Storage management; Pointers and References, linked structures. **Lab work:** Writing C+/C++ functions to perform practical exercises and implement using the algorithms on arrays, records, string processing, queues, trees, pointers and linked structures.

ENT 301: ENTREPRENEURSHIP STUDIES III (2 units)

The course equips students with the tools and capacity needed to start and grow a successful business venture, to identify and maximise opportunities, and to engage in product development that meets target market needs.

ENT 312: VENTURE CREATION (2 Units C: LH 15; PH 45)

Learning Outcomes

At the end of this course, students, through case study and practical approaches, should be able to:

1. describe the key steps in venture creation;
2. spot opportunities in problems and in high potential sectors regardless of geographical location;
3. state how original products, ideas, and concepts are developed;
4. develop business concept for further incubation or pitching for funding;

5. identify key sources of entrepreneurial finance;
6. implement the requirements for establishing and managing micro and small enterprises;
7. conduct entrepreneurial marketing and e-commerce;
8. apply a wide variety of emerging technological solutions to entrepreneurship; and
9. appreciate why ventures fail due to lack of planning and poor implementation.

Course Contents

Opportunity Identification (Sources of business opportunities in Nigeria, Environmental scanning, Demand and supply gap/unmet needs/market gaps/market research, Unutilised resources, Social and climate conditions, and technology adoption gap). New business development (business planning, market research). Entrepreneurial finance (venture capital, equity finance, microfinance, personal savings, small business investment organisations, and business plan competition). Entrepreneurial marketing and e-commerce (Principles of marketing, customer acquisition & retention, B2B, C2C and B2C models of e-commerce, first mover advantage, e-commerce business models and successful e-commerce companies,). Small business management/family business: Leadership & Management, basic bookkeeping, nature of family business and family business growth model. Negotiation and business communication (Strategy and tactics of negotiation/bargaining, traditional and modern business communication methods). Opportunity discovery demonstrations (business idea generation presentations, business idea contest, brainstorming sessions, idea pitching). Technological solutions (the concept of market/customer solution, customer solution, and emerging technologies, business applications of new technologies- Artificial Intelligence (AI), Virtual/Mixed Reality (VR), Internet of Things (IoT), Blockchain, Cloud Computing, renewable energy, etc. digital business and e-commerce strategies).

SEN 304: Software Testing & Quality Assurance (2 Units C: LH 15; PH 45)

Learning Outcomes

At the end of this Course, students should be able to:
1. state the critical importance of software testing in ensuring software quality;
2. explain the difference between validation and verification and their different techniques;
3. describe the concept of quality assurance and differentiate between process assurance and product assurance;
4. describe the different statistical approaches to quality control.

Course Contents

The importance of Software Testing. Understanding Verification and Validation. How to assure it and verify it, and the need for a culture of quality. Avoidance of errors and other quality problems. Inspections and reviews. Testing, verification and validation techniques. Process assurance vs. Product assurance. Quality process standards. Product and process assurance. Problem analysis and reporting. Statistical approaches to quality control.

Lab WorK: Debugging tools; unit testing – black box and white testing techniques; integration and system testing tools; other testing tools – performance testing, load testing, stress testing, regression testing, security testing; manual testing vs automated testing.

SEN 306: Software Construction (2 Units C: LH 15; PH 45)

Learning Outcomes
At the end of this Course, students should be able to:
1. explain the importance of Software Construction and the key construction decisions;
2. describe the key issues in design including key design concepts, levels of design and Abstract Data Types (ADTs);
3. discuss best practices in dealing with routines, fundamental data types and different types of statements; and
4. describe how to ensure software quality through developer testing, debugging and software craftsmanship.

Course Contents
Definition of Software Construction; Its importance; Key construction decisions – choice of programming language, selection of major construction decisions. Design in construction – Key design concepts, levels of design, design heuristics. Abstract Data Types (ADTs). Working Classes. High Quality Routines. The Pseudo Code Programming Process. Fundamental Data Types – Numbers, Characters and Strings, Boolean Variables, Arrays, Tables. Types of Statements – Straight Line Code, Loops, Control Structures; Developer Testing and Debugging. Software Craftsmanship – Layout and Style, Documentation, Personal Character.

Lab Work: Practicals on the most common tools to ensure good software construction. The features include static code analysers to check that code follows coding conventions, special code searching and editing, collaboration support to allow multiple programmers working simultaneously, support for proper code documentation. Practice with IDEs (such as Visual Study Code, NetBeans and Eclipse) on debugging, compilation, running of code, auto completion and version control.

SEN 322: Software Engineering Innovation and New Technology (2 Units C: LH 15)
Learning Outcomes
At the end of this course, students should be able to:
1. explain business models;
2. identify some entrepreneurial opportunities available in Software Engineering;
3. describe business plan and business startup process;
4. explain business feasibility and strategy;
5. explain marketing strategies; and
6. discuss business ethics and legal issues.

Course Contents
Software entrepreneurial process. Principles of software business ownership. Identifying software market opportunities. Entrepreneurial software marketing. Software business communication and negotiation techniques. Feasibility analysis. Entrepreneurial financing. Legal issues. Software business plan development. Risk management.

CLU-SEG 325 Cryptography and Computer Security (3 Units: (Compulsory): 15 H (Lecture), 45H (Practical)

Senate Approved Relevance

This course will provide a standard knowledge of the activities of attackers on communication channels over computer networks, mobile network, IoT networks, Electronic Payment channels etc. It also introduces several techniques to solve or combat such problems in real life cases. The course is streamlined according to university standards to impact knowledge and develop the society.

Overview

Cryptography is the science of providing security for information in the presence of adversaries using mathematical methods of sending messages in disguised form. It is a cornerstone of the modern electronic security technologies used today to protect valuable information resources on the internet. While encryption is probably the most prominent example of a cryptographic problem, modern cryptography is much more than that. Cryptography provides a secure platform to transfer meaningful information in an unreadable format for an attacker posing over a communication channel to hijack details from a message sender to defraud or cause harm to the receiver or both parties. In order to prevent such occurrence, there are several techniques in practice to secure a communication channel.

Objectives

The objectives of the course are to:

1. introduce the concept and essence of security of computer systems and its contents.
2. describe cryptography and its techniques.
3. explain computer security protocols
4. explain the concepts of public and private key cryptography
5. implement a computer security protocol for different applications

Learning Outcomes

1. Student will be able to comprehend computer security and its features.
2. Student will be able to implement suitable security protocols.
3. Student will be able to classify all kinds of security keys.
4. Student will be able to carry out a sample survey of all the techniques.
5. Student will be able to design a simple application with adequate security against active attacks.

Course Content

Introduction to the concept of security. The need for security. Overview of cryptography and its techniques**.** Conventional encryption principles. Basic symmetric key encryption**.** Onetime password and stream ciphers**.** Block ciphers and modes of operation**.** One-way and trapdoor functions. Pseudo-random bit generators. Pseudo-random functions. Private-key encryption. Key agreement. Public-key encryption. Message authentication**.** Digital signatures. Secret sharing. Key distribution. Protocols.

Minimum Academic Standards

CCMAS Benchmarks: A standard computer laboratory equipped with internet enabled capabilities, an android mobile phone and a simulation laboratory may be necessary. Other NUC MAS requirements.

SEN 399: Students Industrial Work Experience Schemed II (3 Units C: PH 135)

At the end of this training, students should be able to:
1. explain how a typical software engineering firm operates;
2. describe the various assignments carried out and the skills acquired during the SIWES period; and
3. submit a comprehensive report on the knowledge acquired and the experience gained during the exercise.

Course Contents
Students are attached to private and public organisations for a period of three months during the third year session long break with a view to making them acquire additional practical experience in all areas of Software Engineering over and above what is gained in SEN 299. Students are supervised during the training period and shall be expected to keep records designed for the purpose of monitoring their performance. They are also expected to submit a report on the experience gained and defend their reports.

CSC 308: Operating Systems (3 Units C: LH 30; PH 45)
Learning Outcomes
At the end of this course, students should be able to:
1. recognise operating system types and structure;
2. describe OS support for processes and threads;
3. recognize CPU scheduling, synchronization, and deadlocks;
4. resolve OS issues related to synchronization and failure for distributed systems;
5. explain OS support for virtual memory, disk scheduling, I/O, and file systems;
6. identify security and protection issues in computer systems; and
7. use C and Unix commands, examine behaviour and performance of Linux, and develop various system programs under Linux to make use of OS concepts related to process synchronization, shared memory, mailboxes, file systems, etc.

Course Contents
Fundamentals of operating systems design and implementation, history and evolution of operating systems, Types of operating systems; Operating system structures; Process management: processes, threads, CPU scheduling, process synchronization; Memory management and virtual memory; File systems; I/O systems; Security and protection; Distributed systems; Case studies.

Lab work: Practical hands-on engagement to facilitate understanding of the material taught in the course. All the process, memory, file and directory management issues will be demonstrated under the LINUX operating system. Also UNIX commands will be briefly discussed. Alternatively, hands-on exposure may be through the use of operating systems developed for

teaching, like TempOS, Nachos, Xinu or MiniOS. Another possibility is through programming exercises that implement and simulate algorithms taught. Simulation of CPU scheduling algorithms, producer-consumer problem, memory allocation algorithms, file organisation techniques, deadlock algorithms and disk scheduling algorithms.

ENT 302: ENTREPRENEURSHIP SKILLS (2 units)

This is a practical entrepreneurship course. In this two course students are exposed to the ***"Chrisland University Sustainability Project (CLUSP)"*** which has a focus of connecting entrepreneurship teaching and theory into student generated sustainable development products and ideas that solve critical real life problems.

GST 308: COMMUNICATION IN FRENCH II (1 unit)

The course builds on the foundations already laid in GST 208 by expatiating on the communication objectives of getting acquainted with somebody: Introducing oneself, Apologizing, Saying one's name, Asking someone for her/his name. The tasks to be realised in the course amongst others pertain to meeting and greeting people, creating a contact list for the class, discovering French alphabet, identifying some objects using politeness formula.

CLU-SEG 411 Software Engineering 2 (3 Units: (Elective): 45H (Lecture)

Senate Approved Relevance

As the world moves rapidly toward software-driven activities in virtually all facets, building the capacity of Chrisland University students to be able to provide robust software solutions would definitely make them stand out as light among their peers in creating atmosphere for national and economic development through software development.

This course focuses on the pragmatic aspects concerned with the production of software from early inception to post-release activities, such as development processes, requirements analysis, design methodologies, development methodologies, verification and testing, and quality control. We will be focusing on Agile Development and Object Oriented Development Techniques, and many development methodologies. By the end of the course student will know what it takes to be a software engineer/software engineering researcher, understand how projects are planned and the techniques that are used to validate, test, and improve their performance.

Overview

This course addresses more advanced topics in software engineering. Topics include the study of project planning, techniques for data-oriented design, object-oriented design, testing and quality assurance, and computer-aided software engineering.

Objectives

The objectives of the course are to:

1. explain the need for Software Engineering and its origins.
2. explain the basic concepts of SDLC, Software Models, Agile Methodologies.
3. compare the traditional waterfall model to the incremental model, the agile model and other appropriate models as well.
4. explain Software Requirements and how to created software requirement specifications
5. define functional and non-functional requirements
6. explain Software Architecture
7. explain the concepts of building a domain model
8. explain the principles of domain analysis

Learning Outcomes

At the end of this course, students should be able to:

1. explain and apply software engineering concepts spanning the entire software production process.
2. implement different software engineering techniques and methodologies.
3. implement different views of software systems using UML diagrams.
4. describe various project management activities including, software re-engineering, re-use, and software quality assurance methodologies.
5. interpret a given requirements model for a simple software system.
6. construct models of the design of a simple software system that are appropriate for the paradigm used to design it.
7. describe secure coding and defensive coding practices.
8. implement code quality standards using metrics and tools

Course Contents

Software Engineering Lifecycle. Software development methods. Project Planning. Project Initiation. Analysis and Design. Requirements Engineering. Functional and Non-functional Requirements. Software Architecture. Process Oriented Analysis and Design. Data Oriented Analysis and Design. Object Oriented Design and Analysis. Implementation and Maintenance, Testing and Quality Assurance. Change Management. Security by Design. Analysis: Building the Domain Model. Test-driven Implementation.

Minimum Academic Standards

CCMAS Benchmarks: A laboratory/studio equipped with computing resources and tools.

CLU- SEG 413 - Cloud Computing and Internet of Things (3 Units: (Compulsory): 15H (Lecture) 45H (Practical)

Senate Approved Relevance

Cloud computing in combination with the Internet of Things will make fundamental changes to the life of mankind, particularly in how information is managed. The cloud is the only technology that can analyze, store, and access the IoT depending on the deployment model.

Because of the nature of on-demand information, cloud computing with an Internet connection is available on any device at any time. Cloud computing works to improve the efficiency of daily tasks in conjunction with the Internet of Things. Cloud computing is about providing a path for data to reach its destination.

In line with the University's motto of cultivating intellectually radiant students, education on cloud computing and Internet of things gives our students timely exposure to trendy matters in the tech space as well as equips them with the requisite knowledge to implement current technology to solve modern issues. Implementing IoT on cloud provides for cost effective means of controlling our devices remotely. This will prove useful in homes, offices and schools.

As hybrid cloud adoption grows, many companies are realizing its benefits and the need to implement it. Cloud computing will continue to open up new opportunities for IoT for a long time to come. Learning them in tandem and on time will provide the much needed edge for our students to think and work effectively in that space.

Overview

Cloud computing enables users to perform computing tasks using services provided over the Internet. The use of the Internet of Things in conjunction with cloud technologies has become a kind of catalyst, the Internet of Things and cloud computing are related to each other.

Due to the rapid growth of technology, the problem of storing, processing, and accessing large amounts of data has arisen. Great innovation relates to the mutual use of the Internet of Things and cloud technologies. In combination, it will be possible to use powerful processing of sensory data streams and new monitoring services. As an example, sensor data can be uploaded and saved using cloud computing for later use as intelligent monitoring and activation using other devices.

This course provides an overview of the Internet of Things (IoT) and Cloud Computing concepts, infrastructures and capabilities. This will help students gain the necessary knowledge to construct IoT systems and use cloud services for processing and storage of the data produced by the IoT devices. Emphasis will be placed on the architecture and design of IoT systems, the different technologies (wireless/mobile/sensor) governing system implementation and the migration of the data to the Cloud for processing.

Objectives

The objectives of the course are to:

1. introduce the students to IoT and Cloud concepts.
2. introduce the students to IoT and Cloud architectures.
3. help the students develop knowledge and critical understanding of the underlying principles of Cloud Computing and IoT system.

4. help students gain practical experience in the development of Cloud-based IoT systems and exposure to appropriate hardware and software platforms that underpin such development.
5. give the students exposure to appropriate hardware and software platforms that underpin the development of Cloud Computing and IoT development.
6. expose the students to Visualizations used to represent IoT devices.
7. discuss security features to protect data stored in the cloud.

Learning outcomes

By the end of the course, student should be able to:

1. describe the cloud and Internet of Things infrastructures
2. select the appropriate sensors and communication protocols for a specific IoT system.
3. use various cloud technologies to deploy cloud services.
4. put cloud computing components—like virtual machines, web applications, and mobile services— into practice.
5. develop methods for moving data from IoT devices to the cloud.
6. put security measures in place to safeguard cloud-based data.
7. display data produced by the IoT device, using visualization techniques.

Course Contents

Introduction to IoT & Cloud. Trends of Computing. Physical design of IoT. Logical design of IoT. IoT enabling technologies. IoT Levels and deployment. Domain specific IoT. IoT Devices and Sensors. IoT Architectures. IoT communication and protocols. IoT Physical Servers and Cloud Offerings. Cloud Computing Fundamentals. Cloud Computing Architectures. Cloud Types and Services Virtualization and Resource Management. Application of IoT & Cloud. IoT and cloud integration. Application development and cloud processing Security and Privacy for IoT/Cloud Computing.

Minimum Academic Standards

CCMAS Benchmarks: A laboratory equipped with internet enabled capabilities. Other NUC MAS requirements.

SEG 415 Computer Aided Software Engineering (3 Units: (Compulsory): 15H (Lecture) 45H (Practical)

Senate Approved Relevance

Educating students on Computer Aided Software Engineering equips them with the much sought after knowledge of relevance in the field of Software Engineering. It gives students the needed edge which comes with being informed about trending matters. Computer Aided Software Engineering CASE is to automate and streamline various software development activities, such as requirements gathering, design, coding, testing, and maintenance. The ultimate goal of CASE is to improve the quality and efficiency of software development, while reducing the costs and time required to bring software products to market.

In line with the University's mission of producing graduates with creativity and innovation, equipping students with CASE knowledge in software development would enable them to develop innovative software systems that meet customer's deadline without sacrificing quality. With the recent push for online courses, digital lesson plans, and remote learning, Computer Aided Software Engineering is more important than ever. With the knowledge of CASE, student would transverse the realm of Project Management with ease.

Overview

Computer-aided software engineering (CASE) is a set of tools and methodologies aimed at improving the software development process. The main objective of CASE is to automate some of the tasks involved in software development and to provide support for the activities that are performed by software engineers.

CASE tools automate or assist in many software development tasks, such as modeling software requirements, designing software architecture, generating code, managing software testing, and tracking software changes. The ultimate objective of CASE is to reduce the time and effort required to develop high-quality software, minimize the potential for human error, and ensure that software development processes are repeatable, consistent, and reliable.

Overall, the goal of CASE is to improve the quality, efficiency, and productivity of software development, making it possible to deliver high-quality software products in a timely and cost-effective manner.

Objectives

The objectives of the course are to:

1. introduce the students to the concept of Computer Aided Software Engineering.
2. explain the benefits of CASE tools in various software methodologies.
3. explain the automation of repetitive and time-consuming tasks using CASE tools.
4. explain the unique features of CASE tools.

5. discuss how to support team collaboration by providing a centralized repository of software development information, making it easier for team members to communicate, share information, and work together.
6. describe systems analysis and design using CASE tools

Learning outcomes

1. Students will become familiar with a variety of CASE tools, including requirements management tools, modelling tools, code generators, testing tools, and configuration management tools.
2. Students will learn about how CASE tools can be used to support software methodologies.
3. Students will learn how CASE tools can be used to model and visualize software systems.
4. Students will understand the distinct features of CASE tools.
5. To make the student know various project management frameworks such as Agile, Waterfall, Scrum, etc., and their strengths and limitations.
6. To make Students gain a deep understanding of software project management, including how CASE tools can be used to analyse and design software systems.
7. To enable students to develop critical thinking and problem-solving skills by learning how to analyse software development problems and use CASE tools to solve them.
8. To enable students to develop communication and collaboration skills by working in teams and using CASE tools to collaborate with others on software development projects.

Course Contents

CASE and its Scope. CASE Environment. CASE Support in Software Engineering. Characteristics of CASE Tools. Architecture of a CASE Environment. CASE technology. Benefits of CASE. CASE tool and its scope. Types of CASE Tools. Characteristics of CASE Tools. How organisations use CASE tools. Functional tool classification. Activity-based tool classification. CASE Integration. Reverse Engineering. Structured Analysis and Design with CASE Tools. Next Generation CASE Tool.

Minimum Academic Standards

CCMAS Benchmarks: A laboratory equipped with internet enabled capabilities. Other NUC MAS requirements.

COS 409: Research Methodology and Technical Report Writing (3 Units C: LH 45)
Learning Outcomes
At the end of this course, students should be able to:

1. describe research, types, approaches, significance of research, research methods, research process, criteria and strategy for good research;
2. discuss the principles of scientific research, scientific investigation, problem formulation, and technique of the research problem;
3. describe the various elicitation methods;
4. develop appropriate data collection instruments;
5. conduct the literature review process; and
6. prepare briefs as well as technical reports and know how to cite referenced works and prepare references and bibliography.

Course Contents

Foundations of Research. Types of Research. Research Approaches. Significance of Research. Research Methods versus Methodology. Research Process. Criteria and Strategy for Good Research. Principles of Scientific Research. Scientific investigation. Problem Formulation and Its Techniques. Developing Research Proposal and Research Plan. Formulation of Research Questions and Hypothesis Testing. Developing Research Proposal and Research Plan. Literature Review. Procedure for Reviewing Related Relevant Studies. Methods for Collection of Primary and Secondary Data. Elicitation Techniques - Questionnaires, Interviewing, Ethnography, etc. Guidelines for Constructing Data Instruments. Methods of AnalysingData in Computing and Related Disciplines.System Design: Architectural design, input design, process design, output design. Use case analysis, sequence diagram, activity diagram, deployment diagram, etc.Types of Reports. Technical Report Writing. Layout and Mechanics of Writing a Research Report. Standard Techniques for Research Documentation. Interpretation and Presentation of Results. How to Cite Referenced Works and Prepare References and Bibliography.

SEN401: Software Configuration Management & Maintenance (2 Units C: LH 15; PH 45)

Learning Outcomes

At the end of this course, students should be able to:
1. state the importance of software configuration management;
2. explain the typical processes in software configuration management; and
3. describe the key issues in software maintenance

Course Contents

Management of the software configuration management process – organisation context for software configuration management, constraints and guidance for software configuration management process. Planning for software configuration management, software configuration management plan, and surveillance of software configuration management. Software configuration identification and software library. Software configuration control – requesting, evaluating and approving software changes, implementing software changes, and deviations and waivers. Software configuration status accounting – software configuration status information and reporting. Software configuration auditing. Key issues in software maintenance – technical issues, management issues, maintenance cost estimation, and software maintenance measurement. Maintenance process – maintenance processes and activities. Techniques for

maintenance – program comprehension, re-engineering, reverse engineering, migration, and retirement.

Lab Work: Practical demonstration of software configuration management processes. Working with software configuration management software. Illustration of software maintenance processes and activities. Working with software maintenance software. Illustration of software re-engineering and reverse engineering techniques.

INS 401 Project Management (2 Units C: LH 30)
Learning Outcomes
At the end of this course, students should be able to:
1. describe project management planning;
2. describe project scheduling;
3. explain management of project resources;
4. discuss project procurement, monitoring and execution; and
5. explain project communication and time management;

Course Contents
Introduction to Project Management. The Project Management Lifecycle. Project management and systems development or acquisition. The project management context, technology and techniques to support the project management lifecycle, and Project management processes. Managing Project Teams: Project team planning, Motivating team members, Leadership, power and conflict in project teams, and Managing global project teams. Managing Project Communication and enhancing team communication. Managing Project Scope: Project initiation, how organisations choose projects, activities, and developing the project charter. Managing Project Scheduling: Common problems in project scheduling, and Techniques for project scheduling. Managing Project Resources: Types of resources (human, capital, time), and techniques for managing resources. Project quality and tools to manage project quality.

SEN 497: Final Year Project I (3 Units C: PH 135)
Learning Outcomes:
At the end of this course, students should be able to:
1. identify researchable project topics in Software Engineering;
2. search and review literature pertinent to identified problem statements;
3. acknowledge and reference sources of information used in the research report;
4. conceptualise and design a research methodology to address an identified problem;
5. determine tools for analysing data collected based on research objectives;
6. write a coherent report on the research conducted;
7. take instruction to accomplish the set goals for the project with the guidance of the research supervisor; and 8. orally present the written project report.

Course Contents
An independent or group investigation to address a Software Engineering problem under the supervision of a lecturer. Before registering, the student must submit a written proposal to the supervisor for review. The proposal should give a brief outline of the project, estimated schedule of completion, and computer resources needed. A formal written report is essential and an oral

presentation may also be required. At the end of the semester, the introduction, literature review and methodology employed should be submitted for grading.

CLU-SEG 408 Advanced Software Testing (3 Units: (Compulsory): 15H (Lecture) 45H (Practical)

Senate Approved Relevance

Educating students on software testing equips them with the much sought after knowledge of relevance in the field of software engineering. It gives students the needed edge which comes with being informed about trending matters. Software Testing has evolved from simple program "check-out" to a broad set of activities that cover the entire software life-cycle. It is the fourth phase of Software Development Life Cycle (SDLC), and about 70% of development time is spent on testing.

In line with the University's mission of showing the light, enlightenment on software testing avails our students with relevant skill to would enable them to develop software systems that are robust and acceptable to all stakeholders.

The relevance of studying software testing to students will vary depending on their individual career goals. However, some potential benefits of including Software Testing in the curriculum to the students include; gaining a better understanding of software development process and the various types of testing methods and levels, as well as developing critical thinking and problem-solving skills.

Overview

Software Testing is the process of evaluating a software application or system to identify defects, errors, or other issues that could affect its performance, functionality, or usability. Software Testing can be performed manually or using automated tools and techniques, and is usually conducted by a team of testers, quality assurance engineers, or software developers.

This course is a structured program designed to build on the fundamentals of software testing covered in Software Testing I. This software testing course may cover more advanced techniques for testing different types of software, such as web applications, as well as more specialized testing methods like security testing or performance testing.

The course will cover topics such as automated testing, performance testing, security testing, and test management. The focus of the course is to prepare students for a career in software testing by providing them with a deeper understanding of testing principles, methods, and tools.

Objectives

The objectives of the course are to:

1. understand and use automated testing frameworks to improve software testing efficiency and effectiveness.
2. learn how to design, execute, and analyze performance tests.
3. develop skills in black box and white box testing techniques.
4. learn how to create and execute effective test plans and test cases.
5. learn how to design and implement test strategies for different types of software.
6. learn how to test web-based systems.
7. learn how to identify and eliminate security vulnerabilities in software using effective security testing techniques.
8. learn how to effectively manage the testing process, including test case prioritization and bug reporting.

Learning outcomes

1. Understand the benefits and limitations of automated testing.
2. Understand the principles and objectives of security testing.
3. Analyze the complexity and coverage of a software system by using test point analysis.
4. Understand the principles and best practices of website testing.
5. Gain understanding of test case design, planning, and execution.
6. Identify and analyse the requirements and risks of a software project to create a test plan.
7. Design and implement effective test cases.
8. Learn the basic principles of test automation and how it can improve the efficiency and accuracy of testing.

Course Contents

The core subjects of Advanced Software Testing to be taught are given below:
Introduction to Advanced Software Testing. Overview of the software testing process. The role of testing in the software development lifecycle. Common types of software defects and testing challenges. Automated Testing. Test automation frameworks and tools. Advantages and Disadvantages of Automated Testing. Problems with Manual Testing. Test Planning and Design. Test plan creation and execution. Test case design principles and techniques. Test case prioritization and risk assessment. Security Testing. Introduction to security testing. Common security threats and vulnerabilities. Techniques and tools for security testing. Test Point Analysis: Model. TPA for Case Study. Testing Websites. Functional and Non-functional Testing (Black and White Box Testing).

Minimum Academic Standards

CCMAS Benchmarks: A laboratory equipped with internet enabled capabilities. Other NUC MAS requirements.

SEN 410: Software Architecture and Design (2 Units C: LH 15; PH 45)
Learning Outcomes

At the end of this course, students should be able to:
1. describe design patterns, frameworks and architectures;
2. explain design of distributed systems and component based design; and
3. describe the techniques of designing for qualities such as reliability, performance, safety, security and reusability.

Course Contents
An in-depth look at software design. Continuation of the study of design patterns, frameworks, and architectures. Survey of current middleware architectures. Design of distributed systems using middleware. Component based design. Measurement theory and appropriate use of metrics in design. Designing for quality attributes such as reliability, performance, safety, security, reusability, etc. Measuring internal qualities and complexity of software. Evaluation and evolution of designs.

Lab Work: Practical demonstration of the use of design patterns, frameworks and architectures. Practical simulation of distributed systems. Illustration of component based design. Working with software design software. Use of software metrics measuring software.

CLU-CSC 434 DEEP LEARNING (3 Units: Compulsory): 30H (Lecture) 30H (Practical)

Senate Approved Relevance:
The Deep Learning Specialization is a foundational program that will help you understand the capabilities, challenges, and consequences of deep learning and prepare you to participate in the development of leading-edge AI technology. With a background in machine learning, graduates of computer sciences **c**an get a high-paying job as a Machine Learning Engineer, Data Scientist, NLP Scientist, Business Intelligence Developer, or a Human-Centered Machine Learning Designer and software developer.

O**verview:**
This course is a subset of machine learning, which is essentially a neural network with three or more layers. These neural networks attempt to simulate the behavior of the human brain—albeit far from matching its ability—allowing it to “learn” from large amounts of data. Deep learning utilizes both structured and unstructured data for training. Practical examples of deep learning are Virtual assistants, vision for driverless cars, money laundering, face recognition and many more. Deep Learning uses artificial neural networks to analyze data and make predictions. It has found its application in almost every sector of business. One of the main advantages of deep learning is that it can automatically learn features from the data, which means that it doesn't require the features to be hand-engineered. This is particularly useful for tasks where the features are difficult to define, such as image recognition.

Objectives: On completion of this course the following objectives should have been met:
1.0 Fundamentals of Deep Learning

2.0 Brief History of Machine Learning
3.0 Introducing Artificial Neural Networks
4.0 Machine Learning Algorithms for Classification
5.0 Machine Learning Algorithms for Regression
6.0 Machine Learning Algorithms for Clustering
7.0 Machine Learning Algorithms for Association Rule Learning
8.0 Performance Metrics and Evaluation

Learning Outcomes 1: Fundamentals of Deep Learning

1.1 Artificial Intelligence (AI)
1.2 The Role of AI
1.3 Machine Learning
1.4 Learning representations from data
1.5 Moving from Machine Learning to Deep Learning
1.6 The "Deep" in Deep Learning
1.7 Relationship between AI, ML and DL
1.8 Understanding how Deep Learning works
1.9 Why Deep Learning
1.10 What makes Deep Learning Different
1.11 Knowing when to use Deep Learning
1.12 Applications of Deep Learning
1.13 The promise of AI

Learning Outcomes 2: Brief History of Machine Learning

2.1 Probabilistic Modeling
2.2 Early Neural Networks
2.3 Kernel Methods
2.4 Decision Trees, Random Forest and Gradient Boosting Machines
2.5 Back to Neural Network
2.6 The modern Machine Learning Landscape

Learning Outcomes 3: Fundamentals of Artificial Neural Networks

3.1 What is a Neural Network
3.2 Benefits of Neural Networks
3.3 Research History
3.4 Models of a Neuron
3.5 Types of Activation Functions
3.6 Neural Network Architectures
3.7 Learning Methods in Neural Networks
3.8 Single Layer Neural Network System (Perceptron)
3.9 Applications of Neural Networks

Learning Outcomes 4: Machine Learning Algorithms for Classification

4.1 What is Classification Task
4.2 What is K-Nearest Neighbour
4.3 Explain the KNN Algorithm
4.4 Understanding Decision Boundaries
4.5 What is L-Means Clustering
4.6 To understand K-Means Clustering Algorithm
4.7 To understand Perceptron Algorithm

4.8 Understanding Decision Tree Algorithms
4.9 Understanding Random Forest Algorithms
4.10 Understanding Support Vector Machines

Learning Outcomes 5: Machine Learning Algorithms for Regression
5.1 Understanding Support Vector Regression
5.2 Understanding Decision Tree Regression
5.3 Understanding Support Vector Regression
5.4 Linear Regression
Learning Outcomes 6: Machine Learning Algorithms for Clustering
6.1 Understanding K- Means Clustering
6.2 Understanding Mean-Shift clustering
6.3 Understanding Gaussian Mixture

Learning Outcomes 7: Machine Learning Algorithms for Association Rule Learning
7.1 What is Q-Learning
7.2 What is R Learning
7.3 TD Learning
Learning Outcomes 8: Performance Metrics and Evaluation
In order to evaluate Regression models, these are metrics in detail:
8.1 Understanding of Mean Absolute Error(MAE)
- Mean Absolute Error (MAE),
- Mean Squared Error (MSE),
- Root Mean Squared Error (RMSE),
- R^2 (R-Squared).

8.2 Understanding Mean Squared Error
8.3 Understanding Root Mean

CSC 402: ETHICS AND LEGAL ISSUES IN COMPUTER SCIENCE (2 Units C: LH 30)
Learning Outcomes
At the end of the course, students should be able to:
1. state laws and regulations related to ethics;
2. identify and explain relevant codes of ethics for computing practice;
3. identify social and ethical issues in different areas of computing practice;
4. review real-life ethical cases and be able to develop ethical resolutions and policies;
5. explain the consequences of ignoring and non-compliance with ethical provisions; and
6. develop a sound methodology in resolving ethical conflicts and crisis.

Course Contents
Addresses social, ethical, legal and managerial issues in the application of Computer Science to the information technology industry. Through seminars and case studies, human issues confronting Computer Science graduates will be addressed. Topics include managerial and personal ethics, computer security, privacy, software reliability, personal responsibility for the quality of work, intellectual property, environment and health concerns, and fairness in the workplace.

CLU-CSC 412 BIG DATA (3 Units: (Compulsory): 45H (Lecture)

Senate Approved Relevance

This course is designed to help Chrisland University Students to understand the impact of massive volume of data that are being generated in this new technological era of social media and to be able to capitalize on the advantage that such huge mass of data provides for insight into business through data analytics.

Studying a course like big data would empower the entrepreneurial ability of the students such that they would be able to provide relevant information that give companies competitive advantage thereby boosting business productivity and profitability.

Overview

Big data is a collection of massive and complex data sets and data volume that include the huge quantities of data, data management capabilities, social media analytics and real-time data. It is a data with so large size and complexity that none of traditional data management tools can store it or process it efficiently. Big data is also a data but with huge size.Big data analytics is the process of examining large amounts of data. There exist large amounts of heterogeneous digital data. Big data is about data volume and large data set's measured in terms of terabytes or petabytes. This phenomenon is called Big data.

This course will introduce students to the fundamental concepts of big data including the definition and taxonomy. Various topics such as big data integration, data generation, storage, management, transfer, to analytics with a focus on state-of-the-art technologies and big data stream techniques, visualizations, and applications for big data. Also covered with real-world applications regarding querying big data with SQL and HiveQL, ending with a discussion on the future of big data.

Objectives

The objectives of the course are to:

1. explain big data
2. discuss big data tools
3. explain big data life cycle
4. explain big data techniques
5. describe big data system

6. describe the use of big data in business

Learning Outcomes

At the end of this course, students should be able to:

1. Understand big data and its basic characteristics
2. compare and contrast existing big data processing data platforms and tools.
3. demonstrate how to query big data repositories.
4. explain clustering and data analysis of big data
5. discuss the trends of big data and its future advancement

Course Contents

Introduction to Big Data. Examples of Big data. Types of Big Data. Characteristics of Big Data. Advantages of Big Data Processing. Big Data Life Cycle. Big Data Tools, Techniques, and Systems. Analytical Theories and Methods. Querying Big Data. Big data and privacy. Using big data in business. Hadoop Distributed File System. Hadoop stack for Big Data. Data placement strategies. Big Data Enabling Technologies. Trends in Big data. Future of Big Data

Minimum Academic Standards

CCMAS Benchmarks: A laboratory equipped with internet enabled capabilities and other NUC MAS requirements.

CLU-CYB 430 FAULT TOLERANT COMPUTING (3 Units: (Elective): 15H (Lecture) 45H (Practical)

Senate Approved Relevance

Training of graduates who are highly skilled in the field of computer science and inculcating in them the basis of fault tolerant computing skills will not only affect their products reliable but also maintainable and dependable. The knowledge of fault tolerant computing and its application into a working system will make our graduates most sort after and this will in turned “shine forth the Light” of the university as contained in its mission and vision.

Overview

Fault-tolerant computing is a design techniques that duplicate components or repeated computations enabling uninterrupted operation in response to component failure. System disasters are caused by neglecting the principles of redundancy and failure independence, which are obvious in retrospect. The central theme of this course is to expose students to the use of reliability and availability computations as a means of comparing fault-tolerant designs. Fault Tolerance is necessary in some of the complex digital systems that serve us in areas requiring high reliability, availability, and safety, such as control of air traffic, aircraft, nuclear reactors, space systems, financial transaction systems, telephone and other communication systems, computer networks, the Internet, military systems, office and home computers, and even home appliances.

Objectives:

The objectives of the course are to:

1. Explain the translators;
2. Explain fault-tolerance and safety-critical systems;
3. State top challenges facing the practice of fault-tolerances;
4. Definition of fault-error-failure;
5. States characteristics of faults, fault models, error models;
6. Explain types of redundancy;
7. Explain the concepts of fault detection and fault tolerance;
8. Provide knowledge in concepts of fault prevention and forecasting;
9. Explain fault – tolerant codes;
10. Explain the advantages and disadvantages of various fault-tolerant programming

Learning outcomes

On completion of the course, students should be able to:

1. Explain fault tolerant computing;
2. Understand the concepts of fault detection;
3. Design a dependable systems for mission critical applications
4. Apply the mean time and mean time to repair;
5. Apply at least 3 redundancy schemes;
6. Apply 4 fault tolerance techniques;

7. Design at least 3 algorithm based fault tolerance;
8. State 4 advantages of and 2 disadvantages of redundancy;
9. State at least 4 challenges in fault tolerance.

Course Contents

Introduction to fault-tolerance and safety-critical systems. Top challenges facing the practice of fault-tolerances. Definition of Fault-Error-Failure. Characteristics of faults, fault models, error models. Fail-Safe and Fail-Operate Systems, Introduction to Failure Mode and Effects Analysis (FMEA), Fault-Tree Analysis (FTA), Risk Analysis (RA), Top 5 Challenges in Fault Tolerance. Redundancy. Error Detection. Damage confinement. Error recovery. Fault treatment. Passive HW redundancy and voting. Information redundancy. Bathtub curve. Reliability Block Diagram. Reliability of Series System. Reliability of Parallel System. Stand-by redundancy. M-of-N System. Reliability block diagram. Steady state and transient solution. Markov models of typical systems. Fault tolerance via redundancy; different forms of redundancy. Hardware fault tolerance. Passive hardware redundancy, triple/N modular redundancy, voting techniques, active hardware redundancy. Hybrid techniques. Fault tolerance at processor level. Byzantine failures. Algorithm-based fault tolerance. Fault tolerant networks. Measures of resilience. Common network topologies and their resilience. Software fault tolerance. Single-version fault tolerance. N-version programming; recovery block approach. Exception handling. Software reliability models. what is checkpointing? Checkpointing level. Shared memory systems and checkpointing. Checkpointing in real-time systems.

Minimum Academic Standards

CCMAS Benchmarks: A laboratory/studio equipped with computing resources and tools.

CYB 404: CLOUD COMPUTING SECURITY (2 Units C: LH 30)
Learning Outcomes
At the end of this course, students should be able to:
1. review the concept of cloud, cloud computing, and benefits of the cloud and knowledge of cloud-enabling technologies, virtualisation and multi-tenanting;
2. describe cloud services and service-oriented architectures, and examine the cloud reference model and cloud service models such as IaaS, PaaS and SaaS;
3. state the cloud deployment models of Public, Private, Hybrid and Community clouds and express how to build a cloud, the open standards and open source cloud management tools, architectural best practices and how to design for the cloud;

4. discuss the security in the cloud and how to efficiently secure your cloud, security for cloud computing;
5. apply the economics of the cloud, costs and payment models, have knowledge of when to use the cloud, cloud strategy, standards and the future;
6. analyse data centres, servers, data storage, data centre networking and virtualisation, cloud cube model, cloud threats, threat mitigation, cloud and security risks, real world issues with cloud computing, and cloud security alliance; and
7. distinguish the National Institute of Standards and Technology, Information Assurance Framework, Cloud Audit, Cloud Management Audit/Assurance Programme, Cloud Business Continuity Planning for secured and effective management of the cloud.

Course Contents

Introduction to cloud computing, cloud computing vendors, cloud computing threats, cloud reference model. Cloud-enabling technologies. Services, Service-Oriented Architectures. Cloud service models. Cloud deployment models. Introduction to data centres: servers, data storage, networking and virtualisation. Data centre networking. Introduction to server virtualisation software: VMware VSphere. Virtual machine management: configuration, placement and resource allocation. Power efficiency in virtual data centres. Fault tolerance in virtual data centres**.** The cloud cube model and security for cloud computing. Security in the cloud. Cloud threats, threat mitigation and security risks. Real world issues with cloud computing. Cloud security alliance. National Institute of Standards and Technology, Information Assurance Framework. Cloud audit. Cloud management audit/assurance programme, Cloud business continuity planning. Building a cloud. Architectural best practices: Designing for the cloud. Economics of the cloud. Cloud strategy. Cloud standards and the future. Security of the cloud.

SEN 498: Final Year Project II (3 Units C: PH 135)

Learning Outcomes

Upon completion of the project, students should be able to:
1. demonstrate technical skills in Software Engineering;
2. demonstrate generic transferable skills such as communication and team work;
3. produce a technical report in the chosen project;
4. defend the written project report; and
5. appreciate the art of carrying out a full-fledged research.

Course Contents

This is a continuation of SEN 497. This contains the implementation and the evaluation of the project. A formal written report (chapters 4-5) has to be approved by the supervisor. A final report comprising chapters 1-5 will be submitted to the department for final grading. An oral presentation is required.

6.2 ADDITION AND DELETION OF COURSES

i. A student shall normally be allowed to add or delete course from the list of courses already registered for, provided he is not violating the minimum and maximum work load regulation.
ii. Such requests to delete or add courses shall normally be expected prescribed form obtainable from the Academic Office.
iii. Any student intending to delete or add course adviser before proceeding to complete the firm.
iv. All requests to add courses must be made within the first four weeks of each semester in which the courses are being offered.
v. Requests to delete courses shall normally not be permitted beyond the end of the 9th week of the commencement of lectures in those courses.

6.3 DURATION OF DEGREE COURSES

To qualify for an honour degree, a student shall complete all his degree requirements within the minimum period prescribed, or a period not exceeding one additional year beyond the prescribed minimum duration. Any student who spends more than one extra year over and above the prescribed minimum duration shall only be awarded a PASS DEGREE.

Students who cannot meet all their degree requirements within two years in excess of the minimum duration, shall be required to withdraw from the University without the award of any degree.

6.4 QUALIFYING FOR THE AWARD OF B.SC DEGREE

- To qualify for the award of a degree, a student must meet the following requirements:
- Must have spent not less than the minimum period and not exceed the maximum duration prescribed for the programme (two years over and above the minimum) or

- Must have registered for and passed all CORE, COMPULSORY and ELECTIVE courses prescribed for the degree programme;
- Must have passed not less than the minimum course units specified for the degree;
- Must obtain not less than a CGPA of 1.00 at the end of the programme;
- Must have satisfied all other University requirements as may be prescribed from time to time by Senate;
- Must also have been found worthy in character throughout the duration of his/her studies.

6.5 DRESS CODE

INTRODUCTION

(i) All Students must dress corporate except when they are in their Halls of Residence or actively participating in sporting activities or filed practicals. Provocative sportswear which expose private parts of the body that are supposed to be concealed (e.g. Chest, navel and thighs) is prohibited. Examples include bum shorts, spaghetti straps, transparent clothing.

(ii) While maintaining good taste with high morality and modesty, the appearance of every student should be characterized by neatness, purity and simplicity, which must conform with the unique standards and virtues of Chrisland Univeristy.

(iii) Tattooing of any part of the body is highly prohibited. Face caps to lecture halls are not allowed.

Sanctions

First time offender: Strong reprimand
Second time offender: Community service
Third time offender: Rustication for one semester

6.5.1 CAMPUS DRESS FOR FEMALE STUDENTS

The following regulations apply at all times to all female dressing on campus.

Dressing which is sexually provocative or expose vital parts of the body that are supposed to be concealed (e.g. chest, navel, and thigh) is prohibited.

The identity of all students must be visible at all times (i.e. faces must be fully visible). The form of dressing that obscures easy identification are not allowed.

(i) Skirts must be long enough to cover the knees and must not be slit above the knees.

(ii) Transparent, sheer, or tight fitting blouses, sweaters or shirts are not permitted.

(iii) Dresses with low necklines or dresses that do not fully cover the shoulders and armpits are not permitted.

(iv) Shorts, slacks and similar attire should not be worn outside the Residence Halls on campus, except while one field trips.

(v) Trouser suits may be worn but with camisole that are not with low necklines.

(vi) Excessive facial make-up, multi coloured nail-polish, artificial nails and bleaching are not allowed.

(vii) Students are allowed moderate use of unnatural braided hair or attachment, which in any case should not exceed neck-length.

(viii) Unnatural braided hair or attachment should match with the natural hair colour. The hair should be well groomed (neat and tidy) all the time. Use of jewelry should be moderate.

(ix) Dangling and bogus earrings are not allowed. Noisy and bogus shoes should be avoided.

6.5.2 CAMPUS DRESSING FOR MALE STUDENTS

(i) Men's dresses for lectures/ and University occasion should consist of trouser (not jeans), shirts and tie to fit.

(ii) Men should not go bare-footed or without shirts anywhere on campus and must wear proper sports uniforms when participating in sports activities or outdoor games.

(iii) Men should not wear muscle tops, earrings or necklaces.

(iv) Men's hair should be kept short, neat and well combed. Beards should be clean shaved.

6.6 RULES AND REGULATIONS GOVERNING THE CONDUCT OF EXAMINATIONS ELGIBILITY

All students who are registered for courses in a given Semester are eligible to sit for examinations in those courses except students in the following categories:

A student who is absent from the University for upwards of three weeks without official permission; such a student shall normally be deemed by Senate to have withdrawn from the University.

A student who fails to attend up to 80% of lectures or practicals in any course.

The implementation of listed above is subject to Senate approval on the recommendation of the Faculty Board.

6.6.1 INSTRUCTIONS TO CANDIDATES

Every candidate shall:

(i) Be admitted into the examination hall only on the production of the University Identity Car and Examination Car.

(ii) Ensure that he/she acquaints himself/herself with and adheres strictly to the instructions governing examinations in the University, including those printed in the front cover of the examination answer booklets.

(iii) Have the full responsibility of ensuring, before the commencement of any examination or test, that nothing incriminating is found on his/her person or on materials he/she legitimately brought into the examination hall (e.g. rulers, four figure tables, etc.).

(iv) Conduct himself/herself in an orderly manner and obey all the instructions of the Invigilator/Examiner.

(v) Not be engaged in, or attempt any other manner of examination malpractice.

(vi) Candidates are not expected in the examination hall earlier than 30 minutes before the commencement of each examination. They are also advised to keep strictly to the seating

arrangements to avoid confusion. Any candidate should not, under any circumstances, remove chairs arranged in halls used for examination purposes.

(vii) Candidates must report at the examination venues punctually at the time scheduled for their papers. A candidate arriving later than 30 minutes after commencement of an examination will be refused entry into the examination hall.

(viii) Candidates are not allowed to leave examination halls prior to 30minutes before the end of the examination.

(ix) Candidates must bring with them to the examination hall their own ink, pen, ruler, erasers and pencils and any materials, which may be permitted by these regulations (as stated hereunder). They are however, not allowed to bring any books or papers, except those that may be permitted or provided by the Department responsible for the examination being held. Candidates are warned in their own interest to ensure that lecture notes, textbooks, jotters, bags or any other unauthorized materials or aids, etc. are not brought into the examination hall.

(x) The Invigilator shall search candidates before they are allowed into the examination hall.

(xi) To ensure orderliness in the examination hall, seats will be arranged according to the Matriculation Number or groups taking examinations at each particular time.

(xii) Candidates are NOT ALLOWED to be in possession of GSM or equivalent handsets in the examination hall. While any examination is in progress, no handset should therefore be found on any candidate.

Communication of any kind between candidates is strictly forbidden during the examination. Any candidate found to be giving or receiving irregular assistance shall face disciplinary action.

(xiii) Silence must be observed in the examination hall. The only permissible way of attracting the attention of the Invigilator is for the candidate to raise his/her hand.

(xiv) The use of scrap paper is not permitted. Rough work must be done in the answer booklets, crossed neatly through and submitted along with the answer booklet/script.

(xv) Candidates offering Mathematics and similar courses must bring their own Mathematical or drawing instruments.

(xvi) Personal copies of Mathematical Tables will not be allowed in examination halls (See Regulation 3 on “The Use of Calculators”).

(xvii) Candidates must use their Matriculation Numbers for the examinations, and not names.

(xviii) Before handing in their answer scripts at the end of the examinations, candidates must satisfy themselves that they have inserted, at the appropriate places, their Matriculation Numbers and the numbers of the questions answered. Except for the question paper and any other materials they may have legitimately brought with them (as indicated above), candidates are not allowed to remove or mutilate any paper or materials supplied by the University.

(xix) Candidates shall not normally be permitted to leave the examination hall during the first and last 30 minutes of any examination.

(xx) At the end of an examination candidates must remain seated while Invigilators go from row to row to collect answers scripts.

(xxi) Candidates are required to sign against their Matriculation Numbers on the Attendance Register at the beginning of the examination and when submitting the answer booklets/scripts.

6.6.2 THE USE OF CALCULATORS

(i) For examination in certain courses, the use of electronic calculators is permitted, but the calculators must be non-programmable. Only one calculator shall be allowed per candidate. Organizers are not permitted.

(ii) The calculator must be small (hand-held) and battery-operated.

(iii) A candidate must not borrow another candidate's calculator during an examination.

(iv) The responsibility for correct operation of the calculator rests with the candidate alone.

(v) Instruction manuals, calculator packets and containers are forbidden in the examination halls. Invigilators and Examinations Assistants shall confiscate them whenever and wherever they are discovered in the examination halls.

(vi) The calculator must be switched off on entry into the examination room and can only be turned on when ready to be used.

(vii) Candidates shall declare their calculators and make them available for inspection by invigilators on entry into the examination hall.

(viii) Contravention of any these regulations shall constitute examination hall.

6.6.3 INSTRUCTIONS TO INVIGILATORS

(i) Invigilators shall normally allow the candidates into the hall ten minutes before commencement of the examination and to sit in an orderly manner in spaces marked according to their Matriculation Numbers. Blank answer booklets and graph papers, where required, shall then be distributed.

(ii) About five minutes before commencement of the examination, the packet of question papers shall be opened and the papers distributed with face downward, while the candidates are warmed not to start until they are told to do so.

(iii) At the scheduled time for the commencement of the examination and after the distribution of papers has ended, the exact time of commencement and the exact time to end the examinations, depending on the periods of time allowed for the various courses scheduled for the hall.

(iv) Invigilators shall pass round the candidates the Attendance Registers corresponding to their courses for the candidates to fill and sign at the beginning and end of the examination.

(v) Invigilators shall familiarize themselves with instructions to candidates and enforce the rules and regulations contained in these instructions.

(vi) Invigilators shall exercise constant and vigilant supervision over the candidates.

(vii) In any case of examination misconduct by candidates, the Chief Invigilator shall require the candidate(s) concerned to write and sign a statement on the incidence, and allow the candidate(s) to proceed with the examination. The Chief Invigilator shall report such

cases to the Dean, the Chief Examiner and Head of Department concerned, within 24 hours.

(viii) Invigilators shall ensure that candidates bring only authorized materials into the Examination Hall. Bags, books and other candidates' properties must be left outside the Examination Hall. For this purpose, Invigilators shall inspect the hall after candidates have been seated before commencement of each examination.

(ix) No candidate may leave the examination hall during the first half-hour of an examination except to go to the toilet or the first-aid room and then only provided that an attendant/Invigilator accompanies him/her.

(x) Normally no candidate shall be admitted after the first half-hour of the examination, and no question paper shall be removed from the hall before the first-hour of the examination has elapsed.

(xi) If any candidate shall have finished his/her paper before the time required, the Chief Invigilator may at his/her discretion allow the candidate to submit his/her answer booklet and retire, except that candidates may not normally leave the examination hall during the first and last half-hour of an examination.

(xii) Invigilators shall ensure that silence is maintained in the examination hall. The only permissible way of attracting the attention of the Invigilator is for the candidate to raise his/her hand.

(xiii) Invigilators must warn candidates of the time, 30 minutes and 5 minutes before the close of an examination.

(xiv) At the close of each examination, Invigilators shall go round to collect from candidates their answer booklets/scripts and check them against the Attendance Register. The candidates shall then sign the Attendance Register and be allowed to leave.

(xv) The Invigilators shall enclose the collected answer booklets/scripts and the corresponding Attendance Registers in the special envelopes provided by the Chief Examiner.

(xvi) The Chief Invigilator shall sign and seal the envelopes and submit them to the Chief Examiner concerned.

6.6.4 ABSENCE FROM EXAMINATION

i. Candidates must present themselves at such University Examinations for which they have registered under these regulations. Candidates who fail to do so for reasons other than illness or accident or other exceptional causes shall be deemed to have failed that examination.

ii. Mis-reading of the time-table and such lapses on the part of the candidates shall not normally be accepted as a satisfactorily explanation for absence.

iii. A student who falls ill during an examination should report in writing to the Dean of his/her college through his/her Head of Department with a report from the University's Director of Medical Services.

iv. A student who is absent from an examination on account of illness confirmed by Medical evidence from the University Medical Centre may obtain the permission of Senate to make up the examination for another occasion, otherwise he/she shall take the regular

examination on the following occasion. Approval for a make-up examination shall be by Senate on the recommendation of the College Board.

6.6.5 EXAMINATION MISCONDUCT TYPES OF EXAMINATION MALPRACTICES

Types of examination malpractices, which are by no means exhaustive, include:

i. Possession of question papers before examination/test or attempt to do so.
ii. Swopping or attempt thereof of answers before, during or after the examination test.
iii. Bringing into the examination hall any unauthorized pieces of paper whatsoever.
iv. Tattooing: inscription of answers, hints or codes thereof on any part of the candidate's body/dress, or in any other manner whatsoever.
v. Passing information from one candidate to another in the examination hall.
vi. Seeking or soliciting any assistance whatsoever from any other student or any other unauthorized person in the examination hall.
vii. Any disorderly conduct before, during or after any examination or test.
viii. Pre-arranged sitting pattern by an student or group of students in the examination hall with a view to facilitating unauthorized exchange of information.
ix. One candidate arranging with another person to write an examination on his/her behalf.
x. Unauthorized possession of the University Answer Booklets or attempts thereof.
xi. Assaulting/manhandling Invigilator and/or Attendant.
xii. Any action or inaction of any student in and around the examination hall, which is inimical to or subversive of the integrity of the University examination process, such as the offences listed above, shall constitute examination misconduct.
xiii. Offering information/assistance and accepting information/ assistance from another student during examination is a bridge of examination regulation and both carry equal punishments.

6.6.6 PROCEDURE FOR HANDLING CASES OF EXAMINATION MISCONDUCT

Examination misconduct shall continue to be processed by the Student Disciplinary Committee.

(i) In any case of examination misconduct by candidates, the Chief Invigilator shall require the candidates concerned to write and sign a statement on the incidence and allow the candidates to proceed with the examination.
(ii) The chief Invigilator shall write his own statement and report the cases of examination misconduct to the Chief Examiner who is also the Head of Department concerned and the Dean of the College.
(iii) On receiving the report, the Dean of the College shall set up a Committee of not less than three members to investigate the case; the report of this Committee shall then be forwarded to the Vice-Chancellor by the Dean of the College.
(iv) At his/her discretion, the Vice Chancellor may act on the report submitted by the Dean of the College or forward it to the Student Disciplinary Committee for advice.

6.6.7 PROCEDURE FOR INVESTIGATING ALLEGED EXAMINATION MALPRACTICE

(i) Failure to observe any of the examination rules of conduct or any breach thereof shall prima-facie constitute examination malpractice or misconduct generally.

(ii) At the discretion of the Chief Invigilator, a candidate may be required to leave the examination room when his/her conduct is judged to be disturbing or likely to disturb the examination. The Chief Invigilator shall submit a written report on any such action to the Head of Department and the Dean of his/her College immediately after the end of the examination.

(iii) Any candidate suspected of any examination irregularities shall be required to submit to the Chief Invigilator a written statement immediately after the paper. Failure to make a written statement shall be regarded as an admission of the allegations against him/her. Reports from Invigilators shall always be through the Chief Invigilators.

(iv) The Dean of the College shall promptly set up a panel of not less than three members of the academic staff to investigate any examination misconduct reported to him/her. The report of such investigation shall be made available, through the Registrar to the Vice Chancellor who shall, on the basis of the recommendations, determine whether or not the matter should receive the attention of the Student Disciplinary Committee.

6.6.8 SANCTIONS FOR PROVENCASES OF EXAMINATION MISCONDUCT

LIST OF OFFENCES

The following are regarded as acts of examination misconduct. [(i) to (xix) are listed as few examples among others];

S/N	Offence	Punishment (Maximum unless otherwise stated)
(i)	Non-display of ID card	Reprimand/Suspension
(ii)	Noise making during an examination	Strong reprimand/Suspension
(iii)	Refusal to submit oneself for search by an Invigilator	Rustication for one Semester
(iv)	Use of scrap papers with written notes in Examination hall	Rustication/Expulsion
(v)	Irregular possession of the University Answer sheets (whether used or unused)	Expulsion
(vi)	Mutilation or removal of any paper or answer script supplied	Expulsion
(vii)	Failure to submit answer script to Invigilator after an examination	Rustication/Expulsion
(viii)	Oral communication between candidates during an examination	Rustication for one semester
(ix)	Communication involving passing of notes or other accessories to aid performance in an examination	Expulsion
(x)	Possession or use of any material such as notes, scraps, etc., to aid performance in an examination	Expulsion
(xi)	Refusal to sit for and/or preventing other students from sitting for an examination	Expulsion
(xii)	Smuggling of prepared answer scripts into an Examination Hall or submission of same under false pretense that they were prepared in the Examination Hall	Expulsion
(xiii)	Use and/or smuggling of cell phones or unauthorized books into the Examination Hall	Expulsion

(xiv)	Possession and/or acquisition of live Question Paper	Expulsion
(xv)	Arrangement/Collusion with another person to write Examination on ones behalf	Expulsion
(xvi)	Writing an examination on behalf of another student	Expulsion
(xvii)	Assault on Invigilator and/or Examination Attendant	Expulsion
(xviii)	Absence from examination without due Authority	To be carried over in the next Semester
(xix)	Any other misconduct concerning, connecting with or pertaining to an examination (not being one of those listed above), which is inimical to or subversive of the integrity of the University examination process, shall be an offence and shall attract appropriate sanctions.	

Sanctions

Any contraventions of any other rules and regulations not listed above shall attract punishment ranging from formal warning to expulsion.